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Rethinking Zombie Firms: From Volatility to Viability

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Rethinking Zombie Firms: From Volatility to Viability

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ABSTRACT: The literature characterizes zombie firms as economically unviable companies that continue to operate. This paper proposes that the majority of zombie firms can be viable, with positive continuation value, and arise endogenously from productivity shocks. Drawing on firm-level data covering the universe of Canadian firms and seven European countries, we document five stylized facts: zombie firms are widespread, unprofitable, unproductive, micro-sized, and transitory. We rationalize these findings by building a general equilibrium model of heterogeneous firms that abstracts from the conventional distortions thought to explain zombie firms. We show that standard productivity shocks are sufficient to explain the empirical patterns, and that the volatility of these shocks can be a key determinant of viable zombie firms, generating shares up to 35%. Since this volatility can also raise aggregate productivity and household utility, the presence of zombie firms is not a sufficient condition for lower welfare. Using a model-based definition to distinguish viability, we find limited evidence—around 0.25%—of unviable zombie firms in the data. Our results challenge the prevailing view that persistently weak firms are unviable and suggest that existing zombie firm measures likely capture viable firms.

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1 Introduction

There are various signs that the process of creative destruction is slowing down across advanced economies. These include: a decline in aggregate productivity growth rates (Ollivaud et al. (2016); Sharpe and Tsang (2018)); an increase in the productivity dispersion between the most and least productive firms (Andrews et al. (2016a); Gu (2019)); an increase in the misallocation of capital between firms (Gopinath et al. (2017); Chen and Tombe (2026)); and a decline in firm entry and exit rates, representing reduced business dynamism (Decker et al. (2014); Amundsen et al. (2026)).

These trends are consistent with the presence of firms that are economically unviable, commonly referred to as 'zombie' firms. The presence of zombie firms, identified by various measures, has been rising over time across developed and developing countries (Adalet McGowan et al. (2018); Banerjee and Hofmann (2018); Banerjee and Hofmann (2020); Albuquerque and Iyer (2024)). Studies suggest that zombie firms are prevalent across industries, hurt the growth prospects of healthy firms, and contribute to lower aggregate productivity growth (Hoshi (2006); Caballero et al. (2008); Adalet McGowan et al. (2018); Banerjee and Hofmann (2018); Acharya et al. (2019); Banerjee and Hofmann (2020); Schmidt et al. (2020); Carreira et al. (2022); Ren et al. (2023); Acharya et al. (2024)).

Yet, two issues remain unresolved in the literature. The first concerns the identification of zombie firms. The literature has relied on alternative approaches: i) using measures of subsidized credit; and ii) using measures of weak performance.[1] While the subsidized-credit measures originate from the seminal studies, the performance-based measures have become popular in the literature due to their lower data requirements, making them feasible for administrative datasets. However, it remains unclear to what extent these measures identify unviable firms (firms with negative continuation value), and whether zombification is a natural part of the lifecycle of viable firms (firms with positive continuation value). This distinction is central because empirical measures rely on observed outcomes rather than continuation values.

The second issue is the lack of a unifying factor to explain zombie firms across countries. The literature has largely attributed the phenomenon to four factors: banks (Peek and Rosengren (2005); Hoshi (2006); Caballero et al. (2008); Giannetti and Simonov (2013); Storz et al. (2017); Acharya et al. (2019); Andrews and Petroulakis (2019); Schivardi et al. (2022)); insolvency regimes (Gouveia and Osterhold (2018); Altman et al. (2024); Kulkarni et al. (2025)); fiscal support (Lam et al. (2017); Qiao and Fei (2022); Amundsen et al. (2025)); and interest rates (Banerjee and Hofmann (2018); Bittner et al. (2021); De Martiis and Peter (2021)). While these factors are relevant in certain countries and time periods, they cannot explain zombie firms at the global level, particularly in countries where banks and insolvency regimes are considered relatively well-functioning, such as in Canada and some European countries.[2]

[1] Some studies have combined the two measures, but for conceptual purposes, we will consider them distinct.

[2] In Canada's case, the banking system is dominated by a small number of large banks and is considered one of the

With these in mind, we propose a parsimonious explanation for zombie firms that reconciles both issues—productivity shocks.[3] These shocks, which occur at the firm or aggregate level, reflect structural volatility arising from market fundamentals. In the absence of distortions, firms that operate under these shocks are viable, since otherwise they would exit, and those distressed by a series of shocks would be misclassified as zombie firms. For the purposes of this paper, we will use the following nomenclature for clarity. *Zombie firms* will refer to those identified by the empirical definitions in the literature. *Viable zombie firms* will refer to the subset that is viable, and thus misidentified relative to the theoretical definition.[4]

The main question is as follows. Can zombie firms arise endogenously without a role for banks, insolvency regimes, fiscal support, or interest rates? Starting with the seminal Hopenhayn (1992), productivity shocks have played a first-order role in models of firm dynamics. In these models, there exists a stationary distribution of heterogeneous firms where a fraction of them enter and exit each period. In every period, firms evaluate their value function, given the realization of the shock, and choose to operate or exit.

We define a zombie firm in the model as having three consecutive years of non-positive profits and age at least ten years old, which closely follows the empirical definition in Adalet McGowan et al. (2018), considered one of the most ubiquitous in the literature. Our novel insight is that the volatility of shocks directly leads to firms experiencing non-positive profits, and therefore the zombie firm share. These dynamics occur even in the Hopenhayn (1992) model. As a proof of concept, we solve the model using standard parameter values and uncover a zombie firm share of 2.5% in partial equilibrium. These zombie firms, although persistently distressed, remain viable, as their expected continuation value is non-negative. Consequently, our main proposition is that productivity shocks can be a key determinant of zombie firms, accounting for their viability and explaining their prevalence globally.

The paper has two main contributions to the literature. The first is empirical. Compared to the literature, we use one of the most comprehensive datasets to date by employing the National Accounts Longitudinal Microdata File from Statistics Canada, which covers the universe of Canadian firms (about 15 million observations) between 2000–2019. This is combined with firm-level data from Moody's (Bureau van Dijk) Orbis for seven European countries (about 25 million observations) between 2004–2019, which includes Belgium, France, Greece, Italy, Spain, Sweden, and the United Kingdom, to provide cross-country coverage.[5]

We focus our empirical findings on five stylized facts for motivation. First, we find that

healthiest systems in the world (Bordo et al. (2015)).

[3]Depending on the model specification, these shocks can represent productivity and/or demand shocks. Given our specification, we refer to them as productivity shocks throughout the paper.

[4]A viable zombie firm would be a misnomer under the theoretical definition since they are unviable by construction. This corresponds to a type II error, as discussed in Caballero et al. (2008), whereby a healthy firm is misclassified as a zombie firm.

[5]Unlike for other countries where Orbis can be relatively well-populated, the coverage for Canada is essentially non-existent outside of using administrative data from Statistics Canada.

zombification is widespread, occurring in all countries in our sample and has fluctuated at the aggregate level, around the GFC period in Canada and the post-GFC period in Europe. Second, the majority of zombie firms are simply unprofitable, rather than a function of interest rates or debt levels, and thus profitability is the key metric behind their identification. Third, a substantial portion of zombie firms have only 1–5 employees, making them micro-sized enterprises. This indicates that any factor behind their existence must be relevant to the smallest firms. Fourth, conditional on survival, about a third of zombie firms recover to a healthy state in any period, highlighting that zombification is transitory. Fifth, zombie firms continue to operate while being about half as productive as healthy firms, implying that even with weak performance, they consider themselves as viable. These findings are all consistent with productivity shocks, since they: occur in all countries; primarily affect profitability; are relevant to firms of all sizes; can lead to recovery; and influence how firms form expectations over future states when assessing their viability.[6]

Rationalizing these findings leads to our second contribution of the paper, which is theoretical. We develop a rich model that incorporates the following features: stationary competitive general equilibrium; static representative household; dynamic heterogeneous firms; endogenous entry and endogenous exit; convex and non-convex capital adjustment costs; and self-financing constraints. The model is purposely built to exclude the role of banks and interest rates by having capital be owned and self-financed by firms.[7] There is no fiscal support, and to mimic a perfectly efficient insolvency regime, there are zero barriers to exit.

The model is solved by discrete value function iteration and has three state variables: productivity, capital, and retained earnings. We do both internal and external calibration for the parameters, and particularly for the shocks, we calibrate the variance of the stochastic shock process to match the share of firms in the data with non-positive earnings. This allows the zombie firm share, which includes the consecutive years criteria and age criteria, to be determined endogenously.[8]

We find that the model closely matches the moments for Canadian and European firms and endogenously reproduces the empirical findings. These include the zombie firm share, the transition matrix between zombie and healthy status, and the size and productivity characteristics of zombie firms, which emerge as untargeted moments. Yet, zombie firms are viable, since firms exit endoge-

[6] As a robustness check, we employ two other zombie firm definitions in addition to Adalet McGowan et al. (2018). First, we use the performance-based measure from Favara et al. (2024), which includes criteria on the interest coverage ratio, sales growth, and leverage. Second, we add a subsidized-credit condition to Adalet McGowan et al. (2018) by requiring that zombie firms pay an average interest rate on their liabilities that is lower than the largest (employees $>$ 500) and healthiest (interest coverage ratio $>$ 10) firms. Under both measures, the empirical findings are robust. This suggests that productivity shocks matter irrespective of how zombie firms are defined.

[7] Since capital is owned by firms, there is no role for interest rates through a rental market. However, firms discount the future, and in other settings this is tied to the interest rate through the household's intertemporal consumption–saving choice. In our setting, since the household is static, it instead represents a time preference.

[8] Both the volatility and the persistence of shocks will influence the share of zombie firms. However, in the absence of volatility, zombie firms will not arise, regardless of persistence. Persistence amplifies zombification at each level of volatility. We therefore focus on the volatility and discipline the persistence using estimates from the data.

nously if their value function is non-positive.[9] For the calibrated variance, our estimates are well in line with those in the literature. The results show that standard productivity shocks are sufficient to explain most zombie firms.

We conduct counterfactual exercises to illustrate the model's dynamics. First, we solve the model for different values of the variance, holding constant the other parameters and maintaining stationary general equilibrium. We find that the variance has a positive monotonic relationship with the zombie firm share, generating shares from 0% to 35%. This demonstrates that the volatility of shocks can account for a broad range of shares, encompassing most estimates in the literature. As the volatility increases, we find a counterintuitive result—aggregate productivity and household utility can rise with zombification. This is due to selection and general equilibrium effects, whereby higher volatility generates both high and low productivity firms, with the least productive firms exiting endogenously. Therefore, even though zombification occurs, the net result can be welfare-improving. The finding indicates that the presence of zombie firms is not a sufficient condition for lower welfare.

We then conduct a negative aggregate productivity shock to simulate a temporary downturn. Ex ante, it is ambiguous whether the zombie firm share should rise, due to weak firms entering zombie status, or fall, due to zombie firms exiting. We find that the share rises but then recovers, consistent with negative shocks inducing firms to enter periods of zombification but not necessarily exit. This occurs simultaneously with an increase in the exit rate. These dynamics match the empirical findings that the share of zombie firms rose and declined around the GFC period in Canada and the post-GFC period in Europe.

Finally, we construct a model-based definition to distinguish viability and identify unviable zombie firms in the data. Our approach is to use the model as a benchmark to create a definition that excludes viable firms. If firms identified by the measure exist in the data, it suggests the presence of factors outside of the model, i.e., the conventional distortions, that allow unviable firms to survive. While various definitions are possible, we settle on a measure that is easy to calculate and includes three criteria: i) firms must have five consecutive years of non-positive earnings; ii) revenue growth is negative between periods $t-4$ and $t-1$ to t; and iii) productivity growth is negative between periods $t-4$ and $t-1$ to t.[10] This measure requires that firms be persistently weak and that they have contracted in both revenue and productivity over the same time period.[11] We show that the measure holds in the model for Canadian and European firms, but when applying it to the data, we estimate the empirical shares at around 0.25%. Consequently, we find limited evidence for the existence of unviable zombie firms.

[9] The model is relatively strict given that capital is self-financed. Some firms would continue to operate if they had access to external borrowing, but instead exit due to liquidity constraints. As financial constraints ease, the state space for viability expands. This implies that the share of viable zombie firms is a lower bound, and would be even higher if temporarily unprofitable firms could borrow against their future earnings from lenders.

[10] We trial alternative specifications for the consecutive years criteria. We find that those based on two and three years, the most common in the literature, are not stringent enough to exclude viable firms.

[11] The criteria for revenue and productivity ensure that growing but persistently unprofitable firms (such as, at one point, Amazon, Uber, Tesla, etc.) are not labeled as zombie firms.

Taken together, this raises important questions about the existence of unviable zombie firms and the applicability of empirical measures used to identify them. Our results challenge the prevailing view that persistently weak firms are unviable, since the continuation value can remain positive even after several consecutive years of unprofitability. Because observed outcomes are not equivalent to economic unviability, existing zombie firm measures likely capture viable firms. The majority of zombie firms can arise endogenously from standard productivity shocks and therefore need not constitute a puzzle. More broadly, the results highlight that market fundamentals imply a 'natural' rate of zombification, and that zombie firms are not a sufficient condition for lower welfare. They also can arise from aggregate downturns while remaining viable. Collectively, these results suggest that firm viability warrants greater attention in future research.

The remainder of the paper is organized as follows. Section 2 discusses the related literature. Section 3 describes the data. Section 4 presents the empirical analysis. Section 5 outlines the model. Section 6 presents the model results. Section 7 concludes.

2 Related Literature

Our paper builds on the large literature surrounding zombie firms, where studies have explored their prevalence and characteristics, as well as their driving factors. The original discussion came from studies that analyzed the Japanese macroeconomic stagnation in the 1990s, which connected banks with zombie firms through evergreening motives and this had adverse effects on other firms and the aggregate economy (Peek and Rosengren (2005); Hoshi (2006); Caballero et al. (2008); Giannetti and Simonov (2013)). Kwon et al. (2015) estimate that annual aggregate productivity growth in Japan would have been 1 percentage point higher had these banks not conducted zombie lending.

The issue of zombie firms was renewed throughout the 2010s as the European sovereign debt crisis unfolded. Studies pointed out that similar zombification was occurring, with banks helping to prop up weak firms in the aftermath of the GFC (Acharya et al. (2019)). Studies such as Storz et al. (2017), Blattner et al. (2019), Andrews and Petroulakis (2019) and Schivardi et al. (2022) connected the presence of weak banks to those of zombie firms, mainly through either poor bank health or stringent capital requirements that led to perverse incentives for maintaining lending relationships.

Until the late-2010s, zombie firms were considered a country-specific or economic-specific phenomenon. Adalet McGowan et al. (2018) and Acharya et al. (2019) showed that zombie firms were widespread across countries and increasing in presence. This sparked concern from policymakers as the zombie firm issue was no longer isolated to specific countries or certain economic periods, but was rather a secular issue that had real economic consequences. Studies suggest that zombie firms crowd out healthy firms in terms of employment, investment, and credit (Banerjee and Hofmann (2018); Andrews and Petroulakis (2019); Acharya et al. (2019)), and that the prevalence of zombie firms is negatively associated with industry and aggregate productivity (Carreira et al. (2022);

Banerjee and Hofmann (2020)).[12]

Relative to these studies, we make several contributions. The first is our use of comprehensive data that covers the universe of Canadian firms. This builds on other studies who have employed similar types of datasets, including Nurmi et al. (2020) for Finland, Carreira et al. (2022) for Portugal, and Cella (2020) for Sweden. An advantage of our dataset is that Canada has a large population of firms and includes information on ownership type (private and listed). Past studies for Canada have focused on listed firms (Banerjee and Hofmann (2020); Grieder and Ortega (2020); Grieder et al. (2021); Altman et al. (2024)), but as Amundsen et al. (2023) show, private zombie firms differ substantially, and thus it is important to include both in the analysis.

On the empirical side, we document five stylized facts regarding zombie firms: they are widespread, unprofitable, unproductive, micro-sized, and transitory. Concerning firm size, many studies focus on medium to large firms, which partly reflects data constraints and the lack of coverage of micro-sized enterprises. This applies to Orbis as well, where Bajgar et al. (2020) show that it underrepresents firms with fewer than ten employees. Our approach is to focus on countries that are relatively well-populated in Orbis and combine the analysis with Canada, which then provides a good balance of cross-country coverage with the comprehensiveness of administrative data. This has the advantage of providing robustness across different data sources.

On the theoretical side, our contribution is related to studies that have explored the factors driving zombification. The literature has largely focused on four factors.[13] Banks, such as evergreening motives and relationship lending (Bruche and Llobet (2014); Storz et al. (2017); Andrews and Petroulakis (2019); Blattner et al. (2019); Acharya et al. (2021a); Acharya et al. (2021b); Chopra et al. (2021); Hu and Varas (2021); Passalacqua et al. (2021); Chari et al. (2021); Chopra et al. (2021); Bonfim et al. (2022); Ozdagli and Wang (2022); Schivardi et al. (2022); Álvarez et al. (2023); Albuquerque and Mao (2025); Hamano et al. (2025)). Insolvency regimes, such as the inability of creditors to force bankruptcy (McGowan et al. (2017); Gouveia and Osterhold (2018); El Ghoul et al. (2021); Becker and Ivashina (2022); Li and Ponticelli (2022); Altman et al. (2024); Kulkarni et al. (2025)). Government support, such as direct and tax-based support, that allows weak firms to

[12]The literature has explored zombie firms in a variety of other avenues. Hong et al. (2021) show that unconventional monetary policy in the form of lower risk premia lead to zombie firms financially restructuring their debt from short-term to long-term, whereas healthy firms respond by increasing investment. Ozdagli and Wang (2022) show using the U.S.–China trade war that firms that rely on bank debt experience a mitigated impact from policy uncertainty shocks, with this effect being stronger for zombie firms. Favara et al. (2024) find that U.S. banks do not engage in zombie lending, while unregulated financial intermediaries do. Havemeister and Horn (2023) find that euro area zombie firms pay higher interest rates, receive less new credit, and do not have financial spillovers on healthy firms. Jordà et al. (2022) find that corporate debt restructuring frictions can lead to more persistent zombie firms over time. De Jonghe et al. (2021) show that banks will reduce zombie lending in sectors in which they have exposure in order to lower the negative spillovers on healthy firms, but this reduction is lessened when the defaults of zombie firms could lower the asset value of healthy firms. Brunner et al. (2024) quantify the impact of product market competition on zombie firms, finding that increased competition reduces the industry asset share of zombie firms and slows their growth. Aretz et al. (2024) show through a creditor–uncertainty channel that the expectation of zombification by healthy firms can have negative real effects.

[13]See Acharya et al. (2022) for an excellent discussion on the first two factors: banks and insolvency regimes.

survive competitive pressures (Lam et al. (2017); Qiao and Fei (2022); Hoshi et al. (2023); Amundsen et al. (2025)). And declining interest rates, which lower incentives to call in non-performing loans and generate higher risk appetite from creditors (Banerjee and Hofmann (2018); Bittner et al. (2021); De Martiis and Peter (2021); Herweg and Kähny (2022)).

We propose a parsimonious explanation for zombie firms that has not been explored in the literature—productivity shocks. Because these shocks are central to market fundamentals and universal across countries, they are a natural candidate to explain zombie firms. Our approach is to combine the literature on zombie firms with that on firm dynamics to highlight how zombie firms can arise endogenously in the absence of distortions. We show that standard productivity shocks are sufficient to explain the empirical patterns and that the volatility of these shocks can be a key determinant of viable zombie firms. Fundamentally, this reveals that persistent weak performance does not necessarily imply economic unviability. Observed zombie firms can reflect dynamics in the lower tail of the productivity distribution, rather than distortions that prevent firm exit or encourage firm survival. Since existing zombie firm measures are based on observed outcomes, their conventional interpretation may need to be reconsidered.

In terms of modeling, our paper is most closely aligned with Faria-e-Castro et al. (2024), but shares features with Acharya et al. (2021b) and Tracey (2025) who also develop models of heterogeneous firms and explore lending to zombie firms.[14] Faria-e-Castro et al. (2024) develop and estimate a firm dynamics model with endogenous entry and exit, and debt as a state variable, which is financed by lenders. When calibrated to U.S. data, they show that evergreening motives can arise endogenously and lead to subsidized lending to less productive firms. Using the zombie firm definition from Favara et al. (2024), they closely match the empirical share. While their results support the evergreening motive, it remains unclear whether zombie firms can exist without the lending channel, among other factors. They also find that subsidized lending leads to larger firms, a result at odds with our finding that a substantial portion of zombie firms are micro-sized enterprises, suggesting that other factors are pertinent. Compared to their model, we develop a different specification. This includes self-financing constraints in lieu of lenders and convex and non-convex capital adjustment costs to match the investment dynamics seen in the data.

Compared to theoretical studies on firm dynamics, our model complements other specifications seen in the literature. These include Hopenhayn and Rogerson (1993) who include labour as a state variable, Hennessy and Whited (2007) who include financial frictions, Clementi and Palazzo (2016) who include capital as a state variable and aggregate fluctuations, Osotimehin and Pappadà (2017) who include financial frictions and aggregate fluctuations, and Midrigan and Xu (2014) and Begenau and Salomao (2019) who include internal and external financing.

[14] Acharya et al. (2021b) show that when firms are hit by large shocks that require unconventional monetary policy, low-capitalization banks lend and evergreen loans to low productivity firms. Tracey (2025) shows that zombie lending drags down aggregate output, investment, and productivity as it prevents the creative destruction of low-quality firms.

3 Data

The Canadian analysis is based on a confidential administrative dataset available at Statistics Canada known as the National Accounts Longitudinal Microdata File (NALMF). This dataset covers all unincorporated and incorporated firms in Canada. It spans a time period of 2000–2019 and includes all industries. Since we require information on the firm's balance sheet and statement of income, we focus on incorporated firms that have filed their T2 corporate tax forms.

For the purposes of this paper, we impose some conditions to clean the data. i) We require firms to be employers by having positive non-missing payroll. This ensures that the dataset is made up of employer firms. ii) Firms must have non-missing tax filing periods of close to a year. This ensures that the observations represent a full year of operation and are comparable to one another. iii) Firms must have non-missing industry. And iv) NAICS 55 (Management of Companies and Enterprises) and NAICS 91 (Public Administration) are excluded. The final sample has roughly 15 million observations and represents the business-sector population of incorporated employer firms in Canada.

For the variables used in the analysis, we define capital as the after-depreciated after-amortized book value of tangible and intangible assets. Age is defined as the difference between the current year and the firm's incorporation date. Earnings is defined as net income before taxes, extraordinary items, and interest payments. Value-added is defined as net income before taxes and extraordinary items + payroll + depreciation and amortization of tangible and intangible assets. Employment is defined as the number of employees. Investment is defined as the net investment in tangible and intangible assets.[15] All variables are deflated when required using deflators from Statistics Canada.

For the European countries, we utilize Moody's (Bureau van Dijk) Orbis, which is a global dataset containing balance sheet and statement of income information on private and listed firms. Since Orbis is not administrative, it can vary over time in terms of coverage and composition. We therefore focus on countries that are relatively well-populated. This includes Belgium, France, Greece, Italy, Spain, Sweden, and the United Kingdom.

We follow Díez et al. (2021) and Kalemli-Özcan et al. (2024) for the data cleaning procedures in Orbis and impose some conditions to align the sample with NALMF. These include removing duplicate observations, applying data quality restrictions for missing and non-positive values, focusing on employer firms and twelve-month annual filings, and excluding NACE 70 (Activities of head offices; management consultancy activities) and NACE 84 (Public administration and defence; compulsory social security). Since the coverage of Orbis is limited prior to the early 2000s, we focus on the period 2004–2019. The final sample has roughly 25 million observations. Most variables align with those in NALMF. However, investment is not available in Orbis, and so we derive it using the perpetual inventory method. All variables are deflated when required using deflators from Eurostat and the World Bank's World Development Indicators.

[15] For results on the investment rate, we exclude the top and bottom 1% to deal with outliers.

3.1 Zombie Firm Definition

The literature has identified zombie firms by two alternative approaches.[16] The first approach is to create a measure of subsidized credit by comparing the firm's interest rate on debt to some benchmark interest rate (Caballero et al. (2008)). Firms are identified to be in zombification if they receive subsidized credit that allows them to remain in the market and survive. Unfortunately, this measure requires detailed information on the firm's debt structure, which makes it unsuitable for many administrative datasets. This has popularized the second approach in the literature, which aims to use indicators of weak performance. Firms that are persistently weak and do not exit are candidates for zombie firms.

Many zombie firm definitions exist under the performance-based measures. These include as defined: Favara et al. (2024), where leverage is above the median, the interest coverage ratio is below one, and the three-year average sales growth is negative; Albuquerque and Iyer (2024), where for two consecutive years the interest coverage ratio is below one, leverage is above the median in the industry, sales growth is negative, and one of the three indicators must be reversed for two consecutive years to exit zombification; Storz et al. (2017), where for two consecutive years the return on assets is negative, net investment is negative, and the debt servicing capacity is below 5%; Banerjee and Hofmann (2018), where the interest coverage ratio is below one for three consecutive years, age is at least ten years old, and Tobin's q is below the median in the industry; and Schivardi et al. (2022), where the three-year moving average of return on assets is below the prime rate (including at least once in the previous two years), and leverage is above 40%.

A common theme across all performance-based measures is their reliance on profitability, usually provided by the interest coverage ratio or return on assets. Studies have shown that the measures tend to be correlated with one another, generating similar levels or trends over time.[17] However, this also implies that the measures could be suffering from the same issue, misidentifying weak viable firms as zombie firms.

For practical analytical and modeling purposes, we focus on the measure from Adalet McGowan et al. (2018). This definition is one of the most ubiquitous in the literature and is provided in Table 1. Zombie firms are defined as having an interest coverage ratio below one for three consecutive years and are at least ten years old. Therefore, zombie firms are mature firms that perform poorly over a period of time without exiting. The definition focuses on mature firms since young firms may have difficulties satisfying their interest payments while they are making investments and expanding their operations.

An advantage of the Adalet McGowan et al. (2018) measure is that it can be easily adjusted to firm dynamics models where interest payments and debt are not applicable. This can be done

[16] See Mingarelli et al. (2022) for an excellent survey on the different zombie firm definitions used throughout the literature.

[17] Amundsen et al. (2025) show among various performance-based measures that they display similar levels and trends for Canada.

Table 1: Zombie Firm Definition

$\frac{\text{EBIT}}{\text{Interest Payments}} < 1$ for three consecutive years & Age $\geq$ 10 years old

Zombie firm definition from Adalet McGowan et al. (2018). EBIT is earnings before interest and taxes. Age is derived from the incorporation date.

by adjusting the earnings criteria, profits in the model, to be less than or equal to zero for three consecutive years.[18] Zombie firms would therefore be making non-positive profits for a period of time and have operated for at least ten years after entry. The age criteria allows time for firms to reallocate their inputs based on their initial draw of state variables and accommodates for convex and non-convex capital adjustment costs that would slow this reallocation down. Since the value function represents the current and discounted expected sum of future profits, the definition implies that even though profits have been non-positive for three consecutive years, the value function remains positive, and thus zombie firms exist. The key question then is understanding why viability remains. Table 2 provides the model-adjusted definition.

Table 2: Model-Adjusted Zombie Firm Definition

Profits ≤ 0 for three consecutive years & Age $\geq$ 10 years old

Profits defined as revenue minus expenses. Age is derived from the entry date.

4 Empirical Analysis

We focus the empirical analysis on five stylized facts regarding zombie firms, which each provide motivation for productivity shocks. The outputs are calculated for the sample of Canadian and European firms.

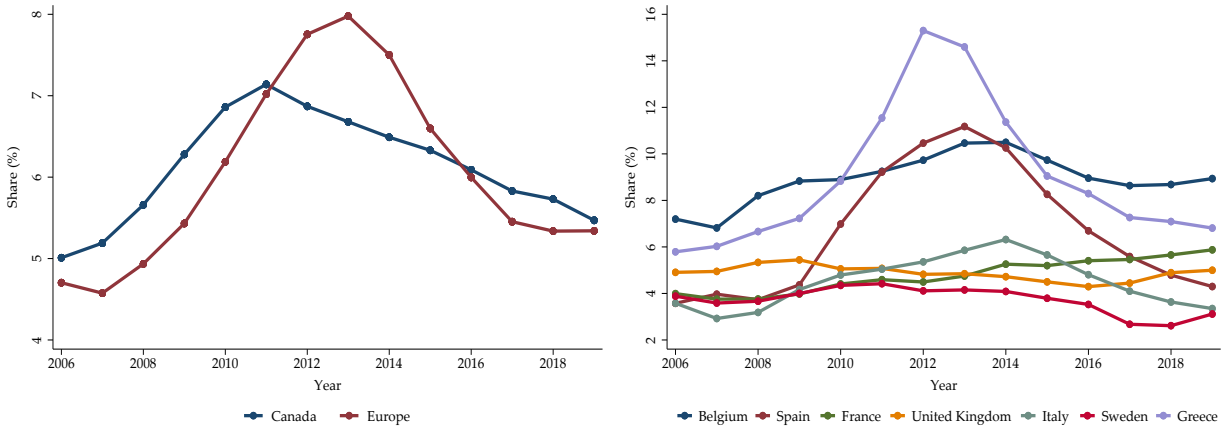
4.1 Prevalence

We begin with the prevalence of zombie firms in Figure 1. We find that zombie firms exist in all countries in our sample. Between Canada and Europe, the shares are roughly equal at the beginning and end of the sample period, where for Canada they rose during the GFC period, while for Europe they rose during the post-GFC period. This is consistent with Canada being relatively more affected by the immediate impact of the GFC due to its closer ties with the United States, while Europe suffered from the so-called 'double-dip' recessions that occurred in 2011–2012 and the associated sovereign debt crisis (Antoshin et al. (2017)). Between the European countries, we find substantial heterogeneity. The shares for Greece, Spain, Belgium, and Italy display a rise throughout the 2010s,

[18] As seen in Figure 2, the majority of zombie firms are identified by non-positive earnings (EBIT ≤ 0) rather than by interest payments.

with Greece and Spain displaying elevated shares above 10% in 2012-2013. This differs from France, Sweden, and the United Kingdom, which have relatively low and flat trends. The finding shows that zombie firms are widespread and that aggregate shocks can induce zombification.

Figure 1: Zombie firms are widespread and fluctuate at the aggregate level

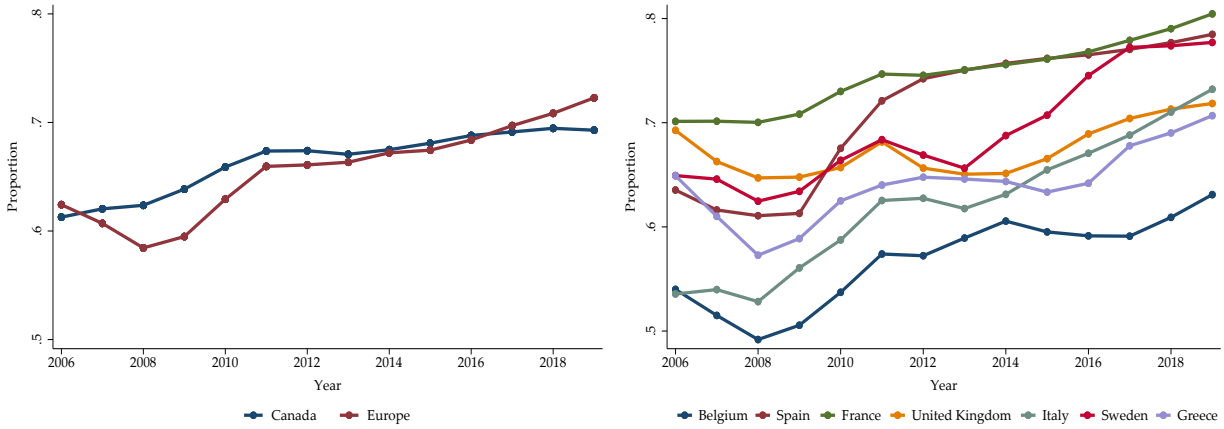


This figure presents the share of zombie firms. The share for Canada is derived from NALMF. The shares for Europe are derived from Orbis. The European share is calculated as the simple average across European countries. Zombie firms are defined as per Adalet McGowan et al. (2018). Last observation is 2019.

4.2 Profitability

The zombie firm definition is satisfied if mature firms cannot meet their interest payments, or if they are simply unprofitable. While the literature has emphasized the former, we find the latter to be relatively more important in identifying zombie firms. In Figure 2, we find that the majority of zombie firms have non-positive earnings (EBIT $\leq$ 0). The proportion ranges from 50% to upwards of 80% across Canada and Europe. This shows that zombie firms are primarily unprofitable irrespective of debt dynamics, i.e., interest rates or debt levels. Given the widespread use of the interest coverage ratio in zombie firm definitions, as well as the return on assets, the results imply that profitability is the key metric behind identification in the literature. We also find that the proportion of non-positive earnings is rising across all countries, indicating that this characteristic is becoming more important over time.

Figure 2: The majority of zombie firms are simply unprofitable



This figure presents the proportion of zombie firms with non-positive earnings (EBIT $\leq$ 0). The proportion for Canada is derived from NALMF. The proportions for Europe are derived from Orbis. The European proportion is calculated as the simple average across European countries. Zombie firms are defined as per Adalet McGowan et al. (2018). Last observation is 2019.

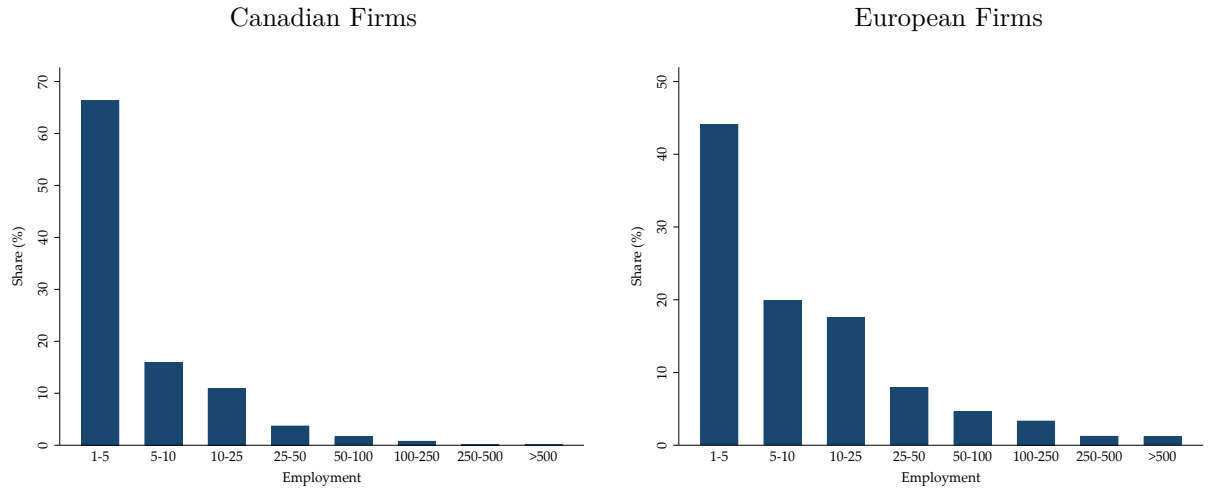
4.3 Firm Size

Zombie firms have generally been regarded as medium to large firms, which then motivates factors such as banks and insolvency regimes. In Figure 3, we highlight an unappreciated characteristic of zombie firms—a substantial portion are actually micro-sized enterprises. About 65% of Canadian zombie firms have between 1–5 employees, while the share is about 45% for European zombie firms. Although the share is lower in Europe, Orbis is known to underrepresent firms with fewer than ten employees (Bajgar et al. (2020)), and so this lower share likely reflects data constraints, implying a lower bound. Nonetheless, the finding indicates that for the factors behind zombie firms, they must be relevant to the smallest firms.

4.4 Persistence

Zombification is generally viewed as a persistent state. However, we find that it is transitory for firms. In Table 3, we present the transition matrix between zombie and healthy status, and the conditional distribution of the number of zombie years, the duration of zombie spells (where spells are defined as consecutive years of zombification), and the number of zombie spells. About one-third of zombie firms will transition to healthy status in any period, indicating that zombification is recoverable. This matches the conditional distribution, where for Canadian (European) zombie firms: 38.49% (41.16%) experience just one year of zombification; 85.26% (90.52%) experience a single zombification spell; and 45.94% (46.20%) experience spells lasting just one year. These findings highlight that it is rare to persistently remain a zombie firm.

Figure 3: A substantial portion of zombie firms are micro-sized enterprises



This figure presents the distribution of zombie firms by employment bins. Zombie firms are defined as per Adalet McGowan et al. (2018).

Table 3: Zombification is transitory

Canadian Firms

	Healthy (%)	Zombie (%)
Healthy (%)	97.03	2.97
Zombie (%)	30.94	69.06

	Number of Zombie Years (%)	Duration of Zombie Spells (%)	Number of Zombie Spells (%)
1	38.49	45.94	85.26
2	23.36	23.98	13.59
3	14.26	12.75	1.15
4	8.66	6.94	.
5	5.50	3.99	.
6	3.43	2.34	.
7	2.21	1.44	.
8	1.40	0.91	.
9	0.93	0.60	.
10	0.60	0.37	.
>10	1.16	0.75	.

European Firms

	Healthy (%)	Zombie (%)
Healthy (%)	97.32	2.68
Zombie (%)	25.75	74.25

	Number of Zombie Years (%)	Duration of Zombie Spells (%)	Number of Zombie Spells (%)
1	41.16	46.20	90.52
2	22.79	22.92	9.10
3	13.41	12.36	0.38
4	8.19	7.11	.
5	5.11	4.19	.
6	3.34	2.65	.
7	2.21	1.68	.
8	1.39	1.04	.
9	0.91	0.69	.
10	0.61	0.47	.
>10	0.89	0.68	.

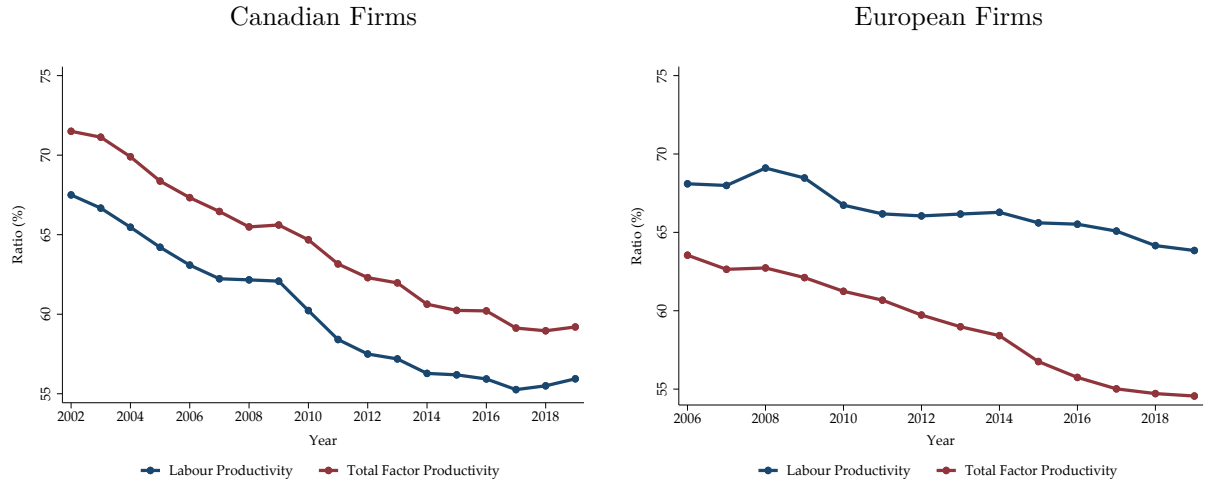
This table presents the transition matrix between zombie and healthy status, and the conditional distribution of the number of zombie years, the duration of zombie spells, and the number of zombie spells. Zombie spells are defined as consecutive years of zombification. Zombie spells greater than three are combined. Zombie firms are defined as per Adalet McGowan et al. (2018).

4.5 Productivity

With zombie firms being largely unprofitable, it implies that these firms are also relatively unproductive. In Figure 4, we present the average productivity of zombie firms relative to healthy firms. We find that zombie firms are close to half as productive as healthy firms by the end of the sample

period, and this has declined over time in both Canada and Europe. This shows that while zombie firms are unproductive, they continue to operate, indicating that they still consider themselves as viable in this weakened state.

Figure 4: Zombie firms operate while being relatively unproductive



This figure presents the average productivity of zombie firms relative to healthy firms. Labour productivity for firm i in year t is defined as $\ln(\text{Value-added}_{it}/\text{Payroll}_{it})$. Total factor productivity is defined as $\ln(\text{Value-added}_{it}) - \alpha_{Ks} \ln\text{Capital}_{it} - \alpha_{Ls} \ln\text{Payroll}_{it}$, where $\alpha_{Ls} = \Sigma_s(\text{Payroll}_{it})/\Sigma_s(\text{Value-added}_{it})$ for sector s and $\alpha_{Ks} = 1 - \alpha_{Ls}$. All variables are in real terms. Zombie firms are defined as per Adalet McGowan et al. (2018). Last observation is 2019.

4.6 Summary

Taken together, the five stylized facts are all consistent with zombie firms facing productivity shocks. These shocks are relevant, at the firm or aggregate level, in all countries, therefore explaining why zombie firms exist globally.[19] These shocks primarily affect profitability, thus generating unprofitable firms irrespective of debt dynamics. These shocks affect firms across the size distribution, and if size and profitability are positively correlated, then micro-sized enterprises would be prone to zombification. Since these shocks are time-varying, a series of shocks can cause firms to enter and recover from zombification, making it transitory for firms. And since these shocks are stochastic, they affect how firms form expectations over future states when assessing their viability, leading to unproductive firms continuing to operate.

4.7 Robustness

These findings have relied on the zombie firm definition from Adalet McGowan et al. (2018). As a robustness check, we replicate the findings using two other measures, one performance-based and

[19] While we focus on variation at the aggregate level, the same logic extends to the industry level. Amundsen et al. (2023) find that the share of zombie firms is highest in commodity-related sectors (mining, oil, and gas), consistent with these firms facing volatile commodity shocks.

one subsidized-credit, to confirm whether the results hold.

The first definition is from Favara et al. (2024). Their measure has three criteria: i) the interest coverage ratio is below one; ii) the three-year average sales growth is negative; and iii) leverage is above the median. We adapt the measure to use revenue instead of sales since it has better coverage in our data, and calculate leverage using total liabilities over assets with the median calculated at the country-year level.

The second definition is based on the subsidized-credit approaches in the literature. We begin with Adalet McGowan et al. (2018) and then augment it by requiring that zombie firms pay an average interest rate on their liabilities that is lower than the median, at the country-year level, of the largest and healthiest firms. These firms are defined as those with more than 500 employees and an interest coverage ratio greater than 10 for three consecutive years. Zombie firms under this measure are a strict subset of Adalet McGowan et al. (2018), and the additional criteria aim to capture the zombie lending channel, i.e., persistently weak firms that receive subsidized credit.

The results for both measures are provided in Appendix A and Appendix B, respectively. Compared to the baseline findings, the results are robust. Zombie firms are found in every country, though the shares under the subsidized-credit measure are much lower at under 3%, indicating that zombie lending is not as prevalent. The majority of zombie firms, above 90%, are identified by having non-positive earnings. A substantial portion, including those receiving subsidized credit, are micro-sized enterprises, and the transition matrices reveal a high probability of recovering in any period of between 36–52%. Lastly, zombie firms are less productive than healthy firms. Overall, irrespective of the definition, the empirical patterns point to productivity shocks as an explanation for zombie firms.

5 Model

In order to rationalize the empirical findings and demonstrate that productivity shocks can generate zombie firms, we turn to a dynamic heterogeneous-firm model.

5.1 Hopenhayn Model

As a proof of concept, we begin with the Hopenhayn (1992) model to show that zombie firms can arise endogenously under a simple setup of partial equilibrium, endogenous entry and exit, and no role for the conventional distortions. Since we employ the standard model, we limit the exposition to a sketch.

5.1.1 Setup

Time is discrete and infinite. There exists a continuum of firms that operate under decreasing returns to scale. In each period, an endogenous mass of firms enter and exit. Operating firms

produce a homogeneous good with one input, labour n. There is no aggregate uncertainty and each firm is subject to a stochastic shock process that determines their productivity draw z. There is a fixed cost to enter f_e and a fixed cost to operate f. Let p denote the output price and w denote the wage, both of which firms take as given. Let the profit function be specified as:

$$\pi(z,p) = \max_{n \geq 0}[pzn^{\alpha} - wn - f], \quad \alpha < 1, \quad f > 0,$$

where firms maximize profit by producing output pzn^{α} and paying the wage bill wn. The aggregate demand function is given by:

$$D(p) = \bar{D}/p, \quad D'(p) < 0,$$

where $\bar{D}$ is a fixed parameter and $D(p)$ declines as p rises. The stochastic shock process is provided by:

$$\ln z' = (1-\rho)\ln \bar{z} + \rho \ln z + \sigma\varepsilon', \quad \rho < 1, \quad \varepsilon' \sim \mathcal{N}(0,1),$$

where productivity evolves according to a stationary auto-regressive Markov process. The value function of an incumbent firm is:

$$v(z,p) = \pi(z,p) + \beta \max\left\{0, \int v(z',p)\, h(dz' \mid z)\right\},$$

where β is the discount factor and $h(dz' \mid z)$ is the probability transition kernel for z. The value function is the cumulative sum of current and discounted expected future profits, where firms at the end of the period choose to exit or stay. Under this specification, firms make their choice before observing z', and thus the expectation is taken within the max operator. The exit threshold at productivity z^* is:

$$\int v(z',p)\, h(dz' \mid z^*) = 0,$$

where firms exit if the expected continuation value is negative, which occurs whenever $z < z^*$. Ex-ante identical potential entrants observe their productivity after paying f_e. Free entry implies that with an entrant probability measure $g(dz)$ and positive mass of entrants m, firms enter until:

$$\beta \int v(z,p)\, g(dz) = f_e,$$

where firms are indifferent between the discounted expected value of entry and its cost. The equilibrium of the model is characterized by the following criteria.

i) Given $\{p, w, \alpha, f, \beta, \bar{z}, \rho, \sigma\}$, incumbents solve:

$$v(z,p) = \pi(z,p) + \beta \max\left\{0, \int v(z',p)\, h(dz' \mid z)\right\}.$$

ii) Given m, the mass of firms is stationary:

$$\mu(dz') = \int \phi(dz' \mid z)\mu(dz) + mg(dz'),$$

where $\mu(dz)$ is the mass measure of incumbent firms and $\phi(dz' \mid z)$ is the probability transition kernel for z conditional on survival.

iii) Given f_e, the free entry condition is satisfied:

$$\beta \int v(z,p)\, g(dz) = f_e.$$

iv) Given $\bar{D}$, the goods market is in equilibrium:

$$\bar{D}/p = \int y(z,p)\, \mu(dz),$$

where $y(z,p)$ is the output of firms.

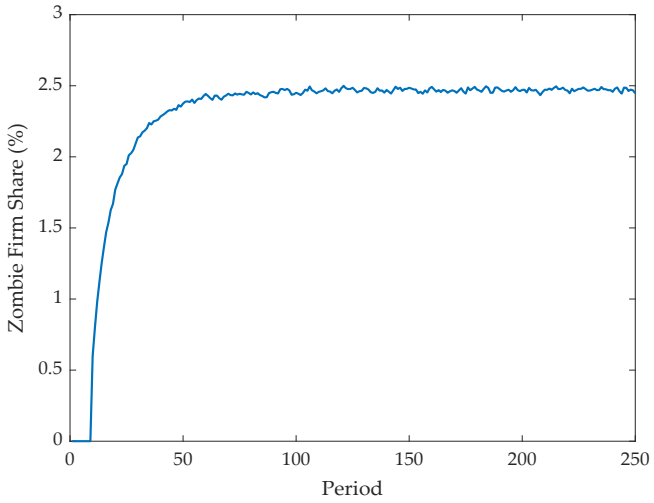
5.1.2 Calibration and Result

We implement the model using Kirkby (2017) and adapt it to calculate the share of zombie firms in partial equilibrium using the model-adjusted definition. The model is solved using standard parameter values: $[w = 1,\ \alpha = 2/3,\ f = 20,\ \beta = 0.8,\ \bar{z} = e^{1.4},\ \rho = 0.9,\ \sigma = 0.2,\ f_e = 40,\ \bar{D} = 100]$.[20] We simulate 100,000 firms for 250 periods and compute the zombie firm share per period. The equilibrium solution is specified by $p^* = 3.6553$ and $m^* = 1.2685$.

The main result is provided in Figure 5. We find that the share converges to a stationary level of 2.5% in partial equilibrium. This highlights that without the conventional distortions, zombie firms can arise endogenously from productivity shocks that induce periods of weak profitability. Nonetheless, these firms do not exit since their expected continuation value is non-negative, and thus they remain viable. The result shows that persistently weak firms will continue to operate.

[20] Although these parameter values correspond to a five-year period ($\beta = 0.8$), we retain them for simplicity. This period length implies a conservative estimate of the zombie firm share.

Figure 5: Viable zombie firms arise endogenously in partial equilibrium



This figure presents the share of zombie firms in the Hopenhayn (1992) model. Zombie firms are defined as having three consecutive years of non-positive profits and age at least ten years old. The model is implemented using Kirkby (2017) and calibrated according to $[w = 1,\ \alpha = 2/3,\ f = 20,\ \beta = 0.8,\ \bar{z} = e^{1.4},\ \rho = 0.9,\ \sigma = 0.2,\ f_e = 40,\ \bar{D} = 100]$.

5.2 General Equilibrium Model

Our result from the Hopenhayn (1992) model motivates a more sophisticated specification that can generate zombie firms from productivity shocks. We develop a rich model with the following characteristics: stationary competitive general equilibrium; static representative household; dynamic heterogeneous firms; endogenous entry and endogenous exit; convex and non-convex capital adjustment costs; and self-financing constraints.

Our aim is to specify a model that includes dynamics in line with the empirical findings, but abstracts from the conventional distortions thought to explain zombie firms, i.e., by having no lenders, no barriers to exit, no fiscal support, and no role for interest rates. Zombie firms, if they arise, will be determined endogenously.

5.2.1 Household

The economy is populated by a static representative household that supplies labour to firms and receives wage income in each period. Firm profits do not accrue to the household and are instead retained by firms. Since the household is static, it does not save, and chooses consumption C and labour supply L_s in each period to maximize its utility function:

$$U(C, L_s) = \frac{C^{1-\kappa}}{1-\kappa} - \frac{1}{\theta^2}\frac{L_s^{1+\gamma}}{1+\gamma},$$

subject to the budget constraint:

$$C = w\, L_s, \quad C > 0,$$

where κ is the coefficient of relative risk aversion, γ is the inverse of the Frisch elasticity, θ is a scaling parameter, and w is the wage. We assume that the household faces a moderate level of risk aversion and a moderate willingness to supply labour ($\kappa = 1$, $\gamma = 1$). The utility function then simplifies to:

$$U(C, L_s) = \log(C) - \frac{1}{\theta^2}\frac{L_s^2}{2}.$$

Taking the first-order condition and substituting the budget constraint yields the following optimal labour supply rule:

$$\frac{\partial U(C, L_s)}{\partial L_s} = w\frac{1}{C} - \frac{1}{\theta^2}L_s = 0,$$

$$L_s^* = \theta^2 \frac{w}{C},$$

$$L_s^* = \theta.$$

The household's optimal supply of labour is constant and determined by the scaling parameter θ. The equilibrium wage w^* adjusts until the labour supplied by the household is equal to the labour demanded by firms. The household's problem is kept simple since the heterogeneity will come from firms.

5.2.2 Incumbent Firms

Time is discrete and infinite. There exists a continuum of firms operating in a perfectly competitive market with three state variables: productivity A, capital K, and retained earnings RE. In each period, an endogenous mass of firms enter and exit. Each firm is subject to a stochastic shock process. There is no aggregate uncertainty.

Firms maximize their value function by choosing labour demand L_d and investment I. The value function is a two-step process, where firms first evaluate whether to exit or stay based on the outturn of each state variable, and conditional on staying, maximize the current and discounted expected value of future profits. Firms exit if the value function is non-positive. Since the exit decision is taken at the beginning of the period, firms in any period are those with a value function that is strictly positive.

$$V(A, K, RE) = \max\left\{0,\ V_{\text{stay}}(A, K, RE)\right\},$$

$$V_{\text{stay}}(A, K, RE) = \max_{L_d,\, I}\left[\pi(A, K, L_d, I) + \beta \int V_{\text{stay}}(A', K', RE')\, h(dA' \mid A)\right],$$

where $\pi(A, K, L_d, I)$ is the profit function, β is the discount factor, and $h(dA' \mid A)$ is the probability transition kernel for A. We assume labour is fully flexible and thus is a choice variable. The value

function is subject to the following laws of motion for A, K, and RE:

$$\begin{aligned}
\ln A' &= \rho \ln A + \varepsilon', \quad A > 0, \quad \varepsilon' \sim \mathcal{N}(0, \sigma^2), \\
K' &= (1-\delta)K + I, \quad K \geq 0, \\
RE' &= RE + \pi(\cdot) - \psi_r I, \quad RE \geq 0.
\end{aligned}$$

Productivity evolves according to a stationary auto-regressive Markov process with persistence parameter ρ and shocks ε', which are normally distributed with mean zero and variance σ^2.[21] Capital evolves where next period's capital K' is made up of depreciated capital $(1-\delta)K$ and investment I. Retained earnings evolve according to an internal financing constraint, where next period's retained earnings RE' is made up of current period's retained earnings RE, profits $\pi(\cdot)$ which increase available financing, and investment I which decreases available financing. Under this specification, firms use their retained earnings, which are built up over time from profits, to finance their investments. The parameter ψ_r reflects the relative price of capital, and accounts for irreversibilities. Its value affects the cost of positive or negative investment, where negative investment (disposing of capital) increases retained earnings.

The law of motion for RE determines the bounds for investment. Firms can invest at feasible points where $RE + \pi(\cdot) \geq \psi_r I$. However, it can occur that there is no feasible I, for example, if profits are highly negative. In such a case, we impose that firms dispose of all their capital and choose the corresponding $I_{\min}$. Consequently, in the next period, they will operate at the lowest level of capital, where they would then evaluate their value function and choose to either exit or stay.

$$I \in \begin{cases} [I_{\min}, I_{\max}] : RE + \pi(\cdot) \geq \psi_r I, \\ [I_{\min}] : RE + \pi(\cdot) < \psi_r I \ \ \forall \ I. \end{cases}$$

The profit function is provided by:

$$\pi(A, K, L_d, I) = p\, A\, K^{\alpha_k} L_d^{\alpha_l} - w L_d - c_f - ac(I, K), \quad \alpha_k + \alpha_l < 1,$$

where p is the perfectly competitive output price, α_k is the output elasticity with respect to capital, α_l is the output elasticity with respect to labour, wL_d is the wage bill, c_f is the fixed operating cost, and $ac(I, K)$ is the adjustment cost function. Firms produce a homogeneous good according to a Cobb-Douglas production function with decreasing returns to scale, where they pay labour costs, an operating cost, and costs associated with adjusting capital. The adjustment cost function

[21] We assume an AR(1) specification for the stochastic shock process since it is the most popular and well-understood in the literature. A different process could raise or lower the share of zombie firms depending on how it is specified. However, as seen in Section 6, given we closely match the calibrated and untargeted moments, the AR(1) appears well-specified for our purpose.

is specified as:

$$ac(I,K) = \psi_{\text{c}} \left(\frac{I}{K}\right)^2 K + \psi_{\text{f}} \, \mathbb{1}\left\{\left|\frac{I}{K}\right| > 0\right\} K,$$

where ψ_c is the convex cost of investment, which reflects that costs become increasingly more expensive for larger investments, and ψ_f is the fixed cost of investment, which reflects inaction due to factors such as indivisible capital. Each cost is scaled by K so that firms cannot grow out of their adjustment costs. The incumbents' problem can be summarized as follows.

i) Given the state variables (A, K, RE), endogenously choose to exit or stay.

ii) If stay is chosen, then choose L_d and I to maximize $V_{\text{stay}}(A, K, RE)$. The choice of L_d follows static optimization, where: $L_d^* = \left(\frac{w}{\alpha_l \, p \, A \, K^{\alpha_k}}\right)^{\frac{1}{\alpha_l - 1}}$.[22]

iii) Enter the next period with (A', K', RE') and repeat the exit decision. In addition to endogenous exit, we assume there is a chance of exogenous exit, denoted by probability ζ.[23]

5.2.3 Entrant Firms

In each period, a mass of potential entrants m consider operating. Potential entrants are endowed with state variables (A_e, K_e, RE_e).[24] We assume that the distribution of A_e equals the invariant distribution of A.[25] The distributions of K_e and RE_e are assumed to be independently and uniformly distributed over the interval $[X_{\min,i}, X_{\max,i}]$ for $i \in \{K, RE\}$, where the density is provided by:

$$f_{X_i}(x) = \frac{1}{X_{\max,i} - X_{\min,i}}, \quad X_{\min,i} \leq x \leq X_{\max,i}.$$

The uniform distribution is chosen because, in implementation, the grids for K and RE are log-spaced. When a uniform distribution is combined with a log-spaced grid, the result resembles a Pareto-like distribution. In order to allow for entrants to differ from incumbents, we apply an exponential tilting function around the mean. The resulting normalized tilted weights for each point

[22] The choice of L_d has dynamic implications given that $\pi(\cdot)$ determines I and RE'. However, the static choice L_d^* remains optimal. This is because since L_d^* maximizes $\pi(\cdot)$, it relaxes the constraint on I and increases its feasible set, which dominates any other choice of L_d. Since L_d^* maximizes $\pi(\cdot)$, RE' is also maximized by its law of motion. As the value function is increasing in RE', this means L_d^* also maximizes $\mathbb{E}[V_{\text{stay}}(A', K', RE')]$. Consequently, L_d^* is both the static and dynamic optimal choice.

[23] This is modeled by using a common uniform draw, $u \sim U[0,1]$, to jointly determine the exogenous exit, $u \leq \zeta$, and the productivity transition, $A' = H(A, u)$, where H is the inverse cumulative distribution function of the transition kernel $h(dA' \mid A)$. Because H is monotonic in u, this has the effect of biasing exogenous exit towards the lower support of A'.

[24] The endowments K_e and RE_e can be interpreted as startup capital provided by owners and shareholders.

[25] In the data, entrants are smaller than incumbents, but they have the same level of average productivity.

x in $[X_{\min,i}, X_{\max,i}]$ are given by:

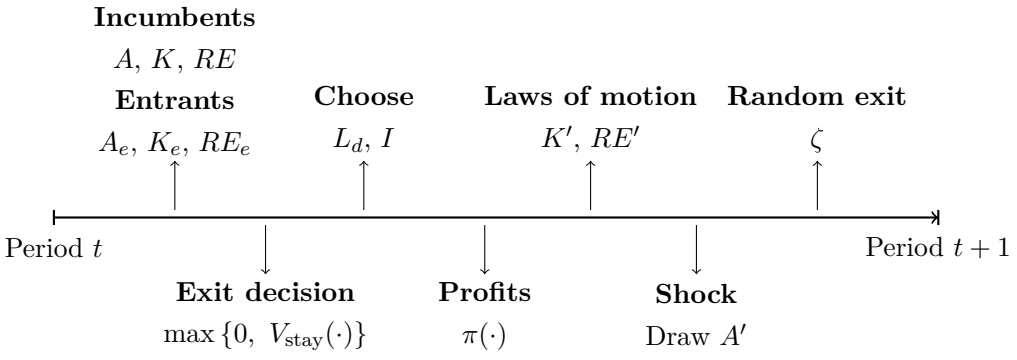
$$w_i(x) = \frac{f_{X_i}(x) \cdot \exp\left(\eta(\tau-1) \cdot \frac{x-\mathbb{E}[X_i]}{X_{\max,i}-X_{\min,i}}\right)}{\int_{X_{\min,i}}^{X_{\max,i}} f_{X_i}(x) \cdot \exp\left(\eta(\tau-1) \cdot \frac{x-\mathbb{E}[X_i]}{X_{\max,i}-X_{\min,i}}\right) dx}, \quad i \in \{K, RE\},$$

where $w_i(x)$ is the normalized weight, η is a scaling parameter, and τ is a parameter that determines the direction and magnitude of the tilt. When $\tau = 1$, the distribution remains the same. When $\tau < 1$, the distribution is tilted towards lower values in the support, and when $\tau > 1$, the distribution is tilted towards higher values in the support. The advantage of this approach is that it preserves the original support while providing flexibility to shape the distributions of K_e and RE_e to match the relative size of entrants to incumbents in the data.

After their initial draw of state variables, potential entrants enter endogenously, choosing to enter if:

$$V(A_e, K_e, RE_e) > 0.$$

The resulting distributions for (A_e, K_e, RE_e) can be organized between those ex ante and ex post. The ex-ante distributions are as previously described, given by the invariant distribution of A and the tilted uniform distributions of K_e and RE_e. The ex-post distributions are conditional on firms with strictly positive value, and therefore will exclude points of (A_e, K_e, RE_e) where this is not satisfied. Since entrants are viable upon entry, they will not exit endogenously until the following period.[26] The timing of the model for both entrants and incumbents is summarized below.



[26] A benefit of this specification, versus one with an entry cost like in Hopenhayn (1992), is that the support for firm viability is identical between entrants and incumbents, both being $V(\cdot) > 0$.

5.2.4 Equilibrium

A stationary competitive general equilibrium is characterized by the following criteria.
(i) Given $\{p, w, \alpha_k, \alpha_l, c_f, \psi_c, \psi_f, \psi_r, \delta, \beta, \rho, \sigma\}$, firms solve:

$$V(A, K, RE) = \max\{0,\ V_{\text{stay}}(A, K, RE)\},$$
$$V_{\text{stay}}(A, K, RE) = \max_{L_d, I}\left[\pi(A, K, L_d, I) + \beta \int V_{\text{stay}}(A', K', RE')\, h(dA' \mid A)\right],$$

such that the laws of motion for (A, K, RE) and constraints on I are satisfied.
(ii) Entry–Exit condition:

$$\underbrace{\int \mathbb{1}\{V(A_e, K_e, RE_e) > 0\}\, \mu_{\text{ent}}(dA_e, dK_e, dRE_e)}_{\text{Endogenous entering firms}} =$$

$$\underbrace{\int \mathbb{1}\{V(A, K, RE) \leq 0\}\, \mu_{\text{end}}(dA, dK, dRE)}_{\text{Endogenous exiting firms}} \quad + \quad \underbrace{\int \zeta\ \mu_{\text{exo}}(dA, dK, dRE)}_{\text{Exogenous exiting firms}},$$

such that the mass of firms entering and exiting are equal, where $\mu(\cdot)$ is the mass measure of firms.
(iii) Stationarity:

$$\mu_{t+1}(S) = \mu_t(S) \quad \forall\ S \subseteq \mathcal{S},$$

such that the distribution of firms across states S in $\mathcal{S}$ is invariant, where $\mathcal{S}$ is the state space (A, K, RE).
(iv) Labour market clearing:

$$\theta = \int \left(\frac{w}{\alpha_l\, p\, A\, K^{\alpha_k}}\right)^{\frac{1}{\alpha_l - 1}} \mu(dA, dK, dRE),$$

such that the labour supplied by the household is equal to the labour demanded by firms.
(v) Goods market clearing:

$$\int (p\, A\, K^{\alpha_k} L_d^{\alpha_l})\, \mu(dA, dK, dRE) = \ C + \int \left[\psi_r I + c_f + ac(I, K)\right]\, \mu(dA, dK, dRE),$$

such that the aggregate supply of goods is equal to the aggregate demand of goods.

5.2.5 Calibration

The model is estimated by internal and external calibration. The calibrated parameters and corresponding data moments are provided in Table 4. All data moments are calculated across the sample period to reflect their secular value, except for the share of non-positive earnings, which

is calculated in the final period.[27] For the European countries, we calculate the moments at the country level and then take the average.

We target eight parameters for the internal calibration. The choice of these are driven by either the equilibrium criteria, feasibility for external calibration, and interest of the paper. The operating cost c_f and the mass of potential entrants m are used to achieve stationary equilibrium, where these are adjusted until we match the average firm turnover rate in the data. Using the operating cost is consistent with the approach in Hopenhayn (1992), and leads to small and unproductive firms being more likely to exit. For m, it is used to oversample the mass of potential entrants such that the mass satisfying $V(\cdot) > 0$ reflects the turnover rate.

Table 4

		Data Moments	
	Parameter	Canadian Firms	European Firms
Internal Calibration			
Turnover Rate	c_f, m	0.0760	0.0960
Relative Size of Entrants to Incumbents	τ	0.3746	0.3491
Standard Deviation of Investment Rates	ψ_c	0.6125	0.6314
Frequency of Zero Investment Rates	ψ_f	0.3179	0.3878
Frequency of Negative Investment Rates	ψ_r	0.0695	0.0653
Share of Non-Positive Earnings	σ	0.2245	0.2274
Wage	w	Endogenous	Endogenous
External Calibration			
Capital Elasticity	α_k	0.2623	0.1974
Labour Elasticity	α_l	0.6763	0.7139
Depreciation Rate	δ	0.0751	0.0625
Discount Factor	β	0.9342	0.9679
Exogenous Exit	ζ	0.0380	0.0480
Productivity Persistence	ρ	0.7640	0.6871
Scaling Parameter for Tilting Function	η	1,000	1,000
Scaling Parameter for Labour Supply	θ	1,000	1,000
Relative Risk Aversion	κ	1	1
Inverse of Frisch Elasticity	γ	1	1
Output Price	p	1	1

Data moments are calculated over the sample period, except for the share of non-positive earnings, which is calculated in 2019.

Since entrants are smaller than incumbents in the data, we use the tilting parameter τ to bias the size of potential entrants until we match the entrant–to–incumbent ratio of median capital. This ensures that firms are smaller upon entry and gives scope for these firms to grow conditional on survival. The adjustment costs, ψ_c, ψ_f, ψ_r, are targeted to match the distribution of investment rates, using ψ_c to match the standard deviation of investment rates, ψ_f to match the frequency of zero investment rates, and ψ_r to match the frequency of negative investment rates.[28] These parameter–moment combinations are standard in the literature (Amundsen (2023)). Since the variance of the stochastic shock process σ^2 is the parameter of interest for the paper, we calibrate its value to match the share of firms in the data with non-positive earnings. As σ^2 increases, firms

[27] The final period is chosen since the zombie firm shares had mostly recovered to pre-GFC levels by the end of the sample period (Figure 1).

[28] Since investment is not available in Orbis and needs to be derived, we adjust the European moment for ψ_f to include the interval $[-5\delta_{ct}, 0]$ using country-year depreciation rates.

receive larger positive and negative shocks, and consequently more firms experience non-positive profits, increasing its share. This feeds into the zombie firm share, which is determined endogenously by the consecutive years criteria and age criteria. Finally, the wage is determined endogenously by equilibrium in the labour market.

The parameters for external calibration are estimated from the data. For the elasticities of the production function, α_k and α_l, we employ the estimation method in Collard-Wexler and De Loecker (2021). This method adjusts for measurement error in capital and provides capital elasticity estimates that are larger and more realistic than other regression-based methods. The estimated elasticities indicate decreasing returns to scale, in line with our model assumptions.[29] The depreciation rate δ is estimated from the weighted average of tangible depreciation and intangible amortization to capital, while the discount factor β is estimated by $1/(1+r)$, where r is the median ratio of interest payments to capital.[30] The exogenous exit probability ζ is assumed to be half of the turnover rate. The persistence parameter ρ is estimated from an AR(1) regression on the log-linear residuals of the production function. The scaling parameter η is set so that τ is estimated at reasonable magnitudes. The scaling parameter θ is set so that firms are distributed between $[X_{\min,i}, X_{\max,i}]$ for $i \in \{K, RE\}$ without unnecessary bunching, as otherwise the policy functions could be distorted. The household parameters, κ and γ, are both set to 1, representing a moderate level of relative risk aversion and a moderate willingness to supply labour. The output price p is normalized to 1, leveraging Walras' law.

Between Canadian and European firms, the data moments highlight several insights on the parameters. The share of non-positive earnings is nearly identical, which reflects the similar zombie firm shares in 2019. European firms display a higher turnover rate, indicating greater business dynamism, while their entrant–to–incumbent ratio is lower, reflecting that incumbents in Europe are larger. There is evidence of irreversibilities and fixed capital adjustment costs, highlighting the importance of including these in the specification. The capital elasticity is larger in Canada, consistent with a bias towards capital-intensive industries, while Europe's labour elasticity is larger, consistent with a bias towards labour-intensive industries. The depreciation rate is lower and the discount factor is higher in Europe, which implies a higher value function at each point in (A, K, RE). While the variance of the stochastic shock process is expected to be similar, the persistence parameter is lower for European firms, indicating that the shocks will have greater mean reversion. These data moments taken together highlight a rich set of dynamics for firms.

[29] Due to missing observations for the TFP estimation procedure, we replace Greece's productivity-related moments with those of Italy.

[30] Although in our specification β represents a time preference, practically, the value needs to be calibrated. The estimates for β are well in line with those in the literature.

5.2.6 Implementation

Since the model has no analytical solution, it is solved numerically by discrete value function iteration (VFI). VFI is chosen as it is a global solution method that is robust to non-convexities, which in our case, come from the capital adjustment costs and internal financing constraints. While VFI is robust, it can be slow, but we mitigate this by leveraging high-performance computing, parallelization, Howard's policy improvement algorithm, and iterative guesses of the initial value function. We set the tolerance on the VFI to be 1×10^{-8}, which is standard in the literature.

The state variables (A, K, RE) are discretized along grid points $[11, 500, 500]$. Productivity A is discretized using the Rouwenhorst method, which Kopecky and Suen (2010) show is the most robust and desirable method versus other approaches. We use log-spaced grids for K and RE. This ensures that there is a greater number of points at lower levels where firms tend to operate. We set the bounds to $[1e{-}6 \leq i \leq 1]$ for $i \in \{K, RE\}$.[31]

Once the model is solved, we simulate 100,000 firms for 250 periods. During the simulation, we use piecewise linear interpolation for the value and investment policy functions so that off-grid points within the bounds are possible.

The eight internal parameters are estimated by an adaptive root-finding algorithm. In calculating the moments, we take the average of the final 50 periods in order to reduce simulation error. The parameters are considered solved if the model's moments fall within 0.5% of the corresponding data moments.

6 Model Results

The estimated parameters and moments are presented in Table 5. We find that the model closely matches the data moments for each of the parameters for Canadian and European firms. This provides support for the model specification and the monotonicity of each parameter–moment combination. In terms of the estimates, the estimate for τ is around 0.65, indicating that the potential entrant distributions are biased towards smaller firms to ensure entrants are smaller than incumbents. The estimate for the convex cost ψ_c indicates that if a firm were to double its capital stock in a year, it would face a cost relative to its capital of 10.05% and 6.93% for Canadian and European firms, respectively. The estimate for the fixed cost ψ_f indicates that a positive or negative investment incurs a cost of 1.34% and 0.55%, respectively. For the estimate of σ, its values are 0.2388 and 0.2254, respectively. These are well in line with those in the literature, which typically range between 0.1–0.3. This provides support that the stochastic shock process, when calibrated to match the share of firms with non-positive earnings, is well-specified.

In Appendix C, we include several figures to highlight the dynamics of the model. These include:

[31] We check that the bounds and grid points align such that no point is farther apart than the depreciation schedule. This ensures that when zero investment occurs, capital appropriately depreciates along the grid.

Table 5

		Canadian Firms			European Firms		
	Parameter	Data	Model	Estimate	Data	Model	Estimate
Internal Calibration							
Turnover Rate	c_f	0.0760	0.0761	0.0003	0.0960	0.0961	0.0008
	m	–	–	84,910	–	–	181,977
Relative Size of Entrants to Incumbents	τ	0.3746	0.3743	0.6826	0.3491	0.3485	0.6308
Standard Deviation of Investment Rates	ψ_c	0.6125	0.6149	0.1005	0.6314	0.6330	0.0693
Frequency of Zero Investment Rates	ψ_f	0.3179	0.3177	0.0134	0.3878	0.3889	0.0055
Frequency of Negative Investment Rates	ψ_r	0.0695	0.0695	1.5142	0.0653	0.0652	2.1387
Share of Non-Positive Earnings	σ	0.2245	0.2247	0.2388	0.2274	0.2278	0.2254
Wage	w	–	–	1.7440	–	–	1.5770

Figure C.1, which presents the ex-ante entrant distributions; Figure C.2, which presents the ex-post entrant distributions; Figure C.3, which presents the count of firms for those starting in the initial period and those entering afterwards; Figure C.4, which presents the simulated moments of the parameters; and Figure C.5, which presents the simulated aggregates of the state and choice variables. All outputs are consistent with the model specification and stationary general equilibrium.

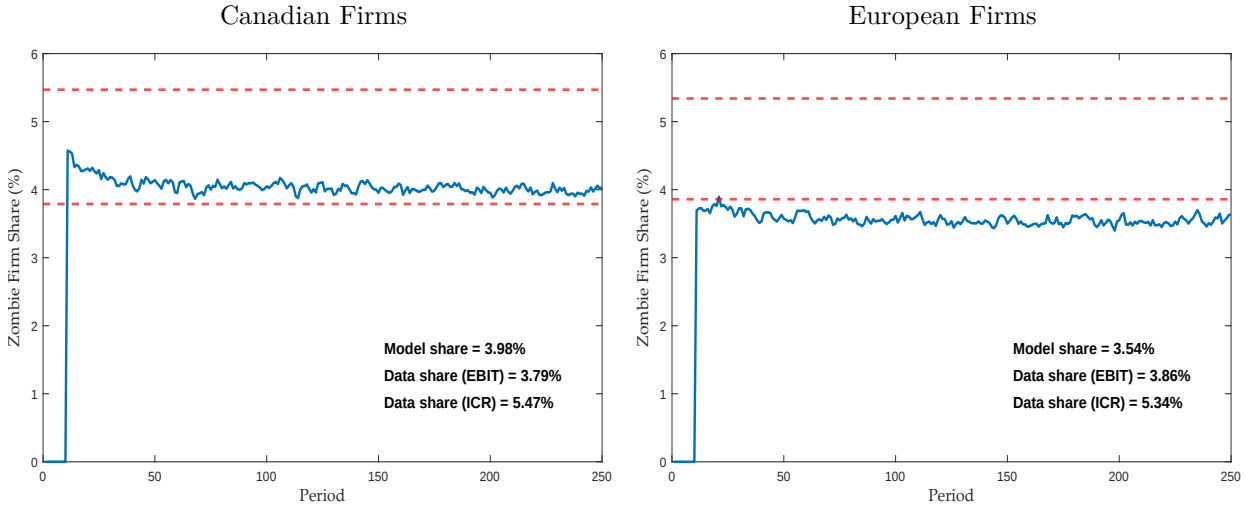
6.1 Viable Zombie Firms

Having calibrated the model, we employ the model-adjusted definition to examine whether zombie firms arise endogenously. The results are presented in Figure 6. We find that the model, in stationary general equilibrium, generates a zombie firm share of 3.98% and 3.54% for Canadian and European firms, respectively. When compared to the empirical shares of 5.47% and 5.34% using the Adalet McGowan et al. (2018) definition, the model accounts for two-thirds of zombie firms. However, when compared to the empirical shares of 3.79% and 3.86% using the non-positive earnings definition, which aligns with the model-adjusted definition and is reflected in Figure 2, the model accounts for virtually all zombie firms. Since firms in the model have a strictly positive value function, these zombie firms are viable. The result shows that zombie firms can arise endogenously from productivity shocks, and this is consistent with zombie firms being prevalent globally.

While the model reproduces the empirical zombie firm share, we find that it also reproduces the transitions into and out of zombification. In Table 6, we present the transition matrix between zombie and healthy status. Despite being untargeted, the model closely follows the empirical transitions. As in Table 3, it is rare to enter zombie status, and a firm has a roughly one-third chance of recovering in any period. This is consistent with firms receiving a series of negative and positive shocks that temporarily push them into a weakened state. The result provides a consistency check on the stochastic shock process and shows that it can realistically generate zombie firms.

So far, the model reproduces three of the empirical findings: that zombie firms are widespread;

Figure 6: The model reproduces the empirical zombie firm share



This figure presents the share of zombie firms in the general equilibrium model. Zombie firms are defined as having three consecutive years of non-positive profits and age at least ten years old. The data share (ICR) is calculated in 2019 and defined as per Adalet McGowan et al. (2018). The data share (EBIT) is adjusted for non-positive earnings.

Table 6: Zombification closely follows the empirical transitions

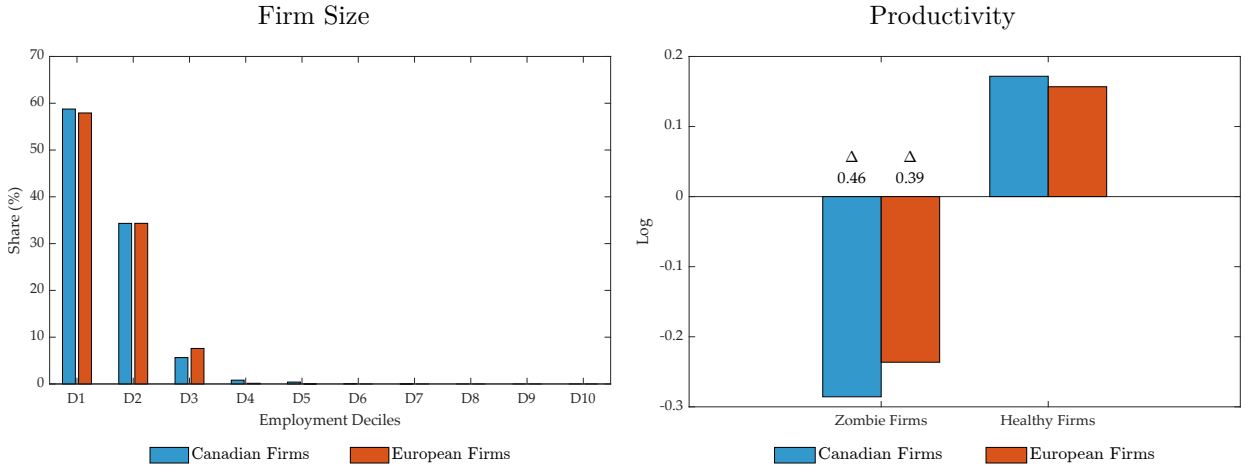
	Canadian Firms		European Firms	
	Healthy (%)	Zombie (%)	Healthy (%)	Zombie (%)
Healthy (%)	98.43	1.57	98.46	1.54
Zombie (%)	24.57	75.43	28.86	71.14

This table presents the transition matrix between zombie and healthy status in the general equilibrium model. Zombie firms are defined as having three consecutive years of non-positive profits and age at least ten years old.

that zombie firms are unprofitable (by construction); and that zombie firms are transitory. The remaining two findings relate to size and productivity. We present the distribution of zombie firms across employment deciles and the average productivity of zombie and healthy firms in Figure 7. We find that zombie firms are smaller than healthy firms, concentrated among the lowest deciles. This is consistent with the empirical finding that a substantial portion of zombie firms are micro-sized enterprises. We also find that zombie firms are less productive than healthy firms, with the log difference in productivity closely matching the data: 0.46 versus 0.42 for Canadian firms, and 0.39 versus 0.40 for European firms.[32] Therefore, while also being untargeted, the model reproduces the size and productivity characteristics of zombie firms. Overall, the model is able to account for the five stylized facts for Canada and Europe, demonstrating that standard productivity shocks are sufficient to explain most zombie firms.

[32]The data values are calculated as the average in 2019 from Figure 4.

Figure 7: Zombie firms are smaller and less productive than healthy firms



This figure presents the distribution of zombie firms across employment deciles and the average productivity of zombie and healthy firms in the general equilibrium model. Zombie firms are defined as having three consecutive years of non-positive profits and age at least ten years old.

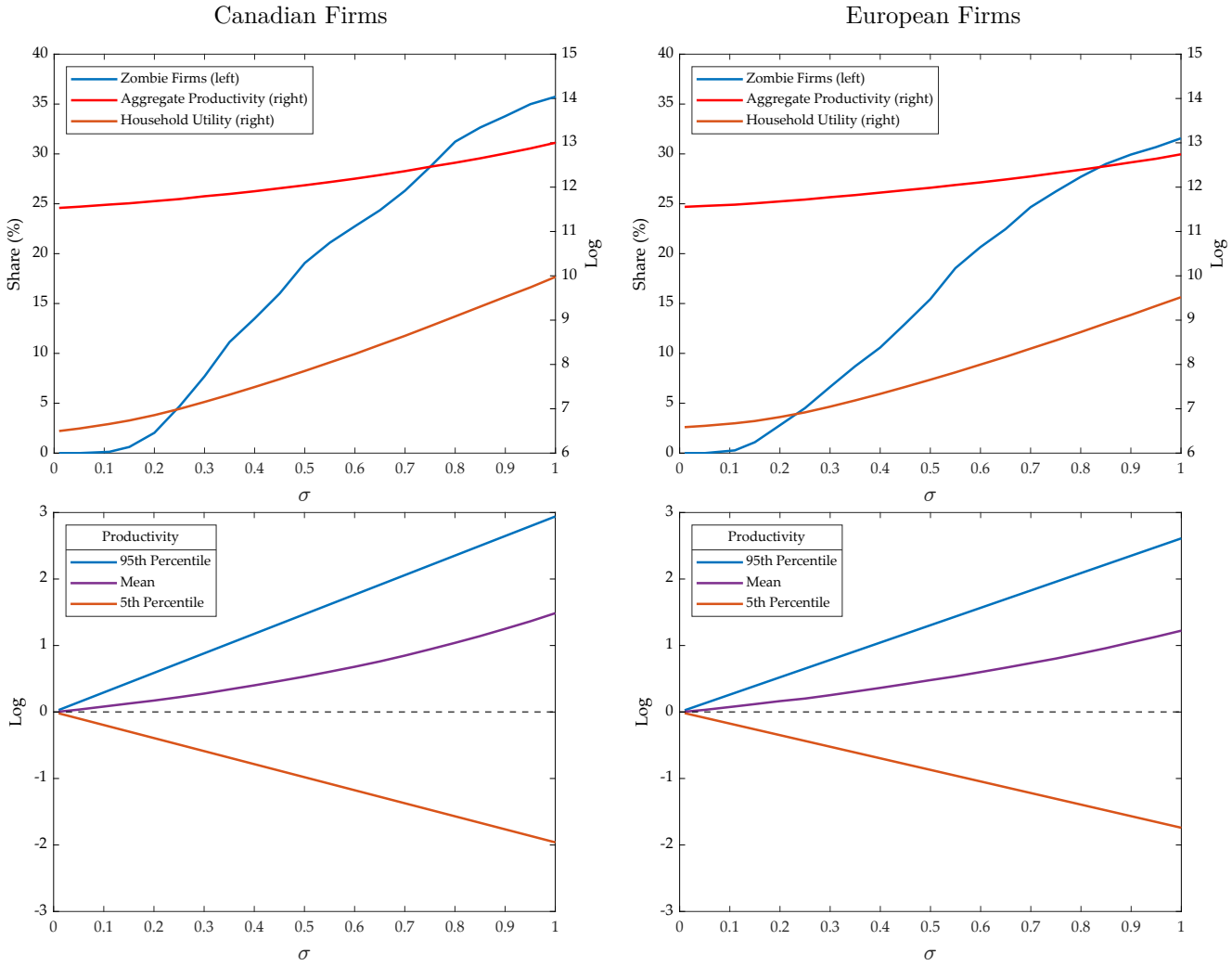
6.2 Counterfactual: Volatility of Shocks

To illustrate how the volatility of shocks can generate zombie firms, we conduct a counterfactual exercise. In each iteration, we solve the model for different values of σ, holding constant the other parameters and maintaining stationary general equilibrium. We then calculate the share of zombie firms, aggregate productivity (unweighted sum), and household utility. The results are presented in Figure 8. We find that the volatility has a positive monotonic relationship with the zombie firm share, generating a broad range of shares from 0% to 35%. This range encompasses most estimates in the literature. When there is no volatility, firms rarely experience non-positive profits since they enter along the state space where the value function is strictly positive, and thus there are no zombie firms. As the volatility increases, firms encounter negative shocks that lead to non-positive profits, of which a subset will experience zombification. The result highlights that differences in the volatility of shocks, reflecting different market fundamentals, can account for cross-country variation in the prevalence of zombie firms.

In combination with the share, we find a counterintuitive result—aggregate productivity and household utility can rise with zombification. This is due to selection and general equilibrium effects. As the volatility increases, the productivity distribution widens, generating both high and low productivity firms. The least productive firms exit endogenously, while the most productive firms continue operating. After accounting for general equilibrium (w^* increases with higher σ), the net result is that while more firms operate in a state of zombification, it is offset by firms in the right tail of the productivity distribution. This is in line with empirical evidence that zombie firms exist in countries that also have 'frontier' or 'superstar' firms, the latter found in Andrews et al. (2016b),

Autor et al. (2020) and Choi et al. (2024). Thus, the presence of zombie firms is not a sufficient condition for lower welfare, and can be the consequence of a process that is welfare-improving.

Figure 8: Higher shock volatility increases zombification and aggregate productivity



This figure presents the share of zombie firms, the level of aggregate productivity (unweighted sum), the level of household utility, and productivity statistics (95th percentile, mean, 5th percentile) for different values of σ of the stochastic shock process. In each iteration, stationary general equilibrium is maintained. Zombie firms are defined as having three consecutive years of non-positive profits and age at least ten years old.

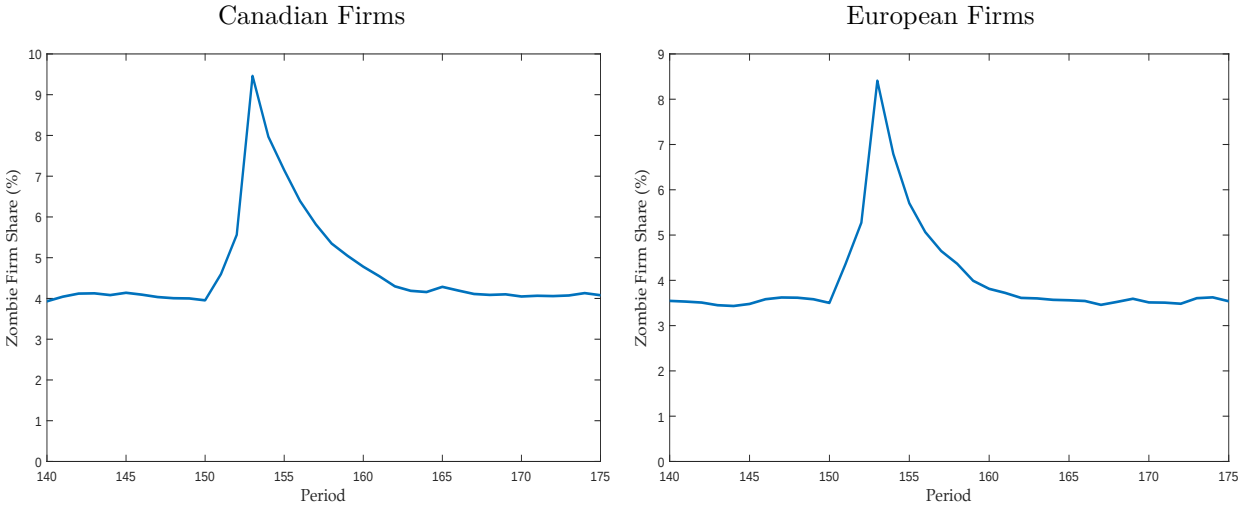
6.3 Counterfactual: Aggregate Shock

Productivity shocks can occur at both the firm and aggregate level, and Figure 1 showed evidence of zombie firms fluctuating around the GFC period in Canada and the post-GFC period in Europe. To examine the impact of an aggregate shock on zombification, we conduct a counterfactual exercise where each firm's productivity is temporarily lowered by one standard deviation σ. The main result

is provided in Figure 9, which presents the zombie firm share, with supporting figures in Figure C.6 and Figure C.7, which present the simulated moments of the parameters and the simulated aggregates of the state and choice variables.

Ex ante, the impact of a negative aggregate shock on the share is ambiguous, since weak firms could enter zombie status, or zombie firms could exit. Further, the literature has argued that the use of the consecutive years criteria is to ensure that the definition is robust to temporary shocks, implying that there should be no impact. We find that the share rises, more than doubling, when firms are hit by a negative aggregate shock, but this recovers as the model returns to equilibrium. This increase in the share is accompanied by an increase in the exit rate (Figure C.6), indicating that while the weakest firms exited, other firms entered zombie status.[33] The result shows that aggregate productivity shocks can induce zombification, and this is consistent with the empirical trends in Canada and Europe around the GFC and post-GFC periods. The pace of recovery is also consistent, where it takes several years (up to a decade) for the initial shock to dissipate. Since firms in the model are viable, the higher incidence of zombification in this scenario is not a failure of creative destruction or evidence of distortions, but reflects a state of temporary weakness across firms, during which some will exit and some will recover as the economy normalizes.

Figure 9: Zombification rises in a temporary downturn



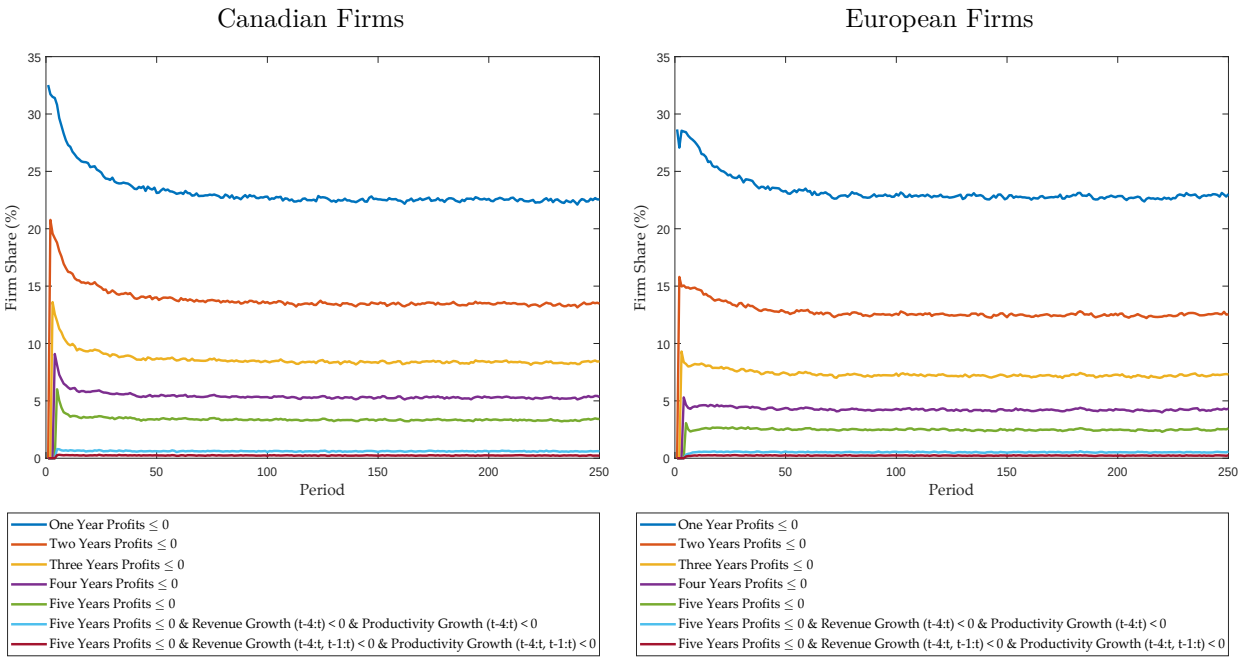
This figure presents the share of zombie firms during an aggregate shock in the general equilibrium model. The shock is simulated in period 150, where each firm's productivity is temporarily lowered by one standard deviation σ. Zombie firms are defined as having three consecutive years of non-positive profits and age at least ten years old.

[33] Between the numerator and denominator of the zombie firm share, the dynamics are driven by the numerator (number of zombie firms), which more than doubles.

6.4 Unviable Zombie Firm Definition

Having shown that most zombie firms can be viable, the question becomes whether unviable zombie firms exist. By leveraging the model, we can distinguish viability and create an appropriate measure. Definitions in the literature typically rely on intuition, whereby measures are constructed with the assumption that they capture unviable firms. However, we can directly test this. In Figure 10, we present the share of firms in the model by different performance criteria. We find that viable firms exist even up to five consecutive years of non-positive profits. At two and three years, the most common criteria used in the literature, the shares equal about 15% and 10%, respectively. Consequently, definitions based on these criteria are likely not stringent enough to exclude viable firms.

Figure 10: Viable firms exist even with persistent non-positive profits



This figure presents the share of firms by different performance criteria in the general equilibrium model.

To distinguish viability, we use the model as a benchmark to construct a definition that excludes viable firms. If firms identified by the measure exist in the data, it suggests the presence of factors outside of the model, i.e., the conventional distortions, that allow unviable firms to survive. While various definitions are possible, we settle on a measure that complements the model specification and is easy to calculate: i) firms must have five consecutive years of non-positive earnings; ii) revenue growth is negative between periods $t-4$ and $t-1$ to t; and iii) productivity growth is negative between periods $t-4$ and $t-1$ to t. As illustrated in Figure 10, this measure excludes virtually

all viable firms.[34] The definition requires that firms be persistently weak and have contracted in both revenue and productivity over the same time period before entering zombie status.[35] The criteria for revenue and productivity are consistent with the idea that a firm with growing revenue or rising productivity, even if unprofitable, should not be classified as a zombie firm.[36] The $t-4$ condition is to accommodate for possible measurement error in the data, while the $t-1$ condition is to ensure that recovering firms do not enter zombie status. In Figure C.8, we trial the definition using alternative consecutive years criteria, which shows that fewer than five consecutive years of non-positive earnings still capture viable firms.

Taking the model-based definition to the data, we find limited evidence for the existence of unviable zombie firms in Figure 11. The shares for both Canada and Europe hover at around 0.25% of firms. For the European countries, we find more heterogeneity, with the share for Spain reaching 1% during the sovereign debt crisis. This is consistent with the literature for this period and suggests that the definition captures zombie firms driven by other factors, when relevant. By the end of the sample period, the shares across the European countries vary between 0.1–0.4%.

In Figure C.9, we present each component of the measure to highlight its contributions. This is split by: the earnings criteria; the earnings and revenue criteria; and the earnings and productivity criteria. We find that each component has a meaningful impact on the combined share, with the earnings criteria alone yielding shares between 1–2% for Canada and Europe, and between 1–4% for the European countries.[37] The additional criteria of revenue and productivity further lower their respective shares to under 1.5% each. Overall, unviable zombie firms are not prevalent, and this is consistent with the majority of zombie firms being viable.

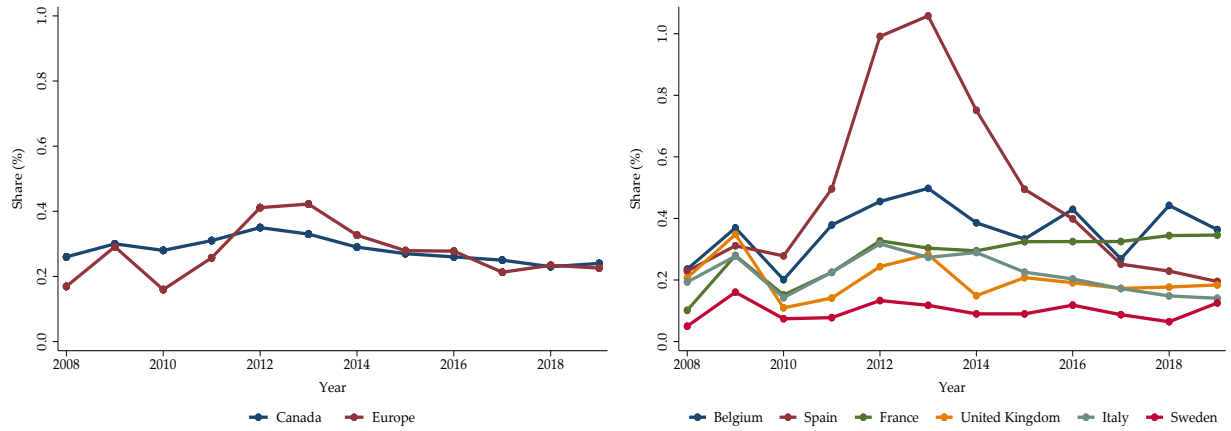
[34] While the definition is constructed to exclude all viable firms in the model, there is a tiny share of firms that remain. Given the model's dynamics, these firms have a combination of productivity, capital, and retained earnings such that they remain viable. However, we would consider these firms as outliers. An appropriate definition should be broadly applicable, i.e., not be so stringent that firms could never satisfy it in the data, and a definition tailored to these firms is at risk of being impractical. We therefore settle on a measure that balances applicability and restrictiveness. Moreover, by being relatively lenient, this implies an upper bound on the unviable zombie firm share in the data, which reinforces our findings.

[35] This decline in revenue and productivity is consistent with empirical evidence that exiting firms tend to be smaller and less productive than surviving firms (Dunne et al. (1989); Griliches and Regev (1995); Foster et al. (2008)).

[36] A firm with growing revenue but declining productivity would be expanding both outputs and inputs, and thus expanding its operations. A firm with declining revenue but rising productivity would be restructuring its inputs and/or innovating, and contributing to aggregate productivity growth. In both cases, there is no reason to consider such firms unviable. This is supported by their existence in the model where they satisfy the non-positive earnings criteria. However, when both revenue and productivity decline, the firm is in a deteriorating position with worsening future prospects, leading to unviability and thus exit in the model.

[37] When comparing these empirical shares to the corresponding share in Figure 10, it implies that these firms are viable.

Figure 11: Unviable zombie firms are almost non-existent in the data



This figure presents the share of unviable zombie firms for Canada and Europe. The share for Canada is derived from NALMF. The shares for Europe are derived from Orbis. The European share is calculated as the simple average across European countries. Unviable zombie firms are defined as having five consecutive years of non-positive earnings (EBIT ≤ 0), negative revenue growth between periods $t-4$ and $t-1$ to t, and negative productivity growth between periods $t-4$ and $t-1$ to t. Greece is excluded due to missing observations for the TFP estimation procedure. Last observation is 2019.

7 Conclusion

This paper proposes that the majority of zombie firms can be viable, with positive continuation value, and arise endogenously from productivity shocks. Drawing on firm-level data covering the universe of Canadian firms and seven European countries, we document five stylized facts: zombie firms are widespread, unprofitable, unproductive, micro-sized, and transitory. These are all consistent with a stochastic shock process generating zombie firms.

We rationalize these findings by first uncovering that viable zombie firms can arise endogenously in the seminal Hopenhayn (1992) model. We then build a general equilibrium model of heterogeneous firms. We show that standard productivity shocks are sufficient to explain the empirical patterns. Using counterfactual exercises, we demonstrate that the volatility of these shocks can be a key determinant of viable zombie firms, generating a broad range of shares from 0% to 35%. This volatility can also raise aggregate productivity and household utility. We then conduct a negative aggregate productivity shock and show that a temporary downturn increases the zombie firm share. This mirrors the empirical trends. Finally, leveraging the model, we construct a model-based definition to distinguish viability and identify unviable zombie firms. However, when applying it to the data, we find limited evidence for their existence—around 0.25%—in Canada and Europe.

Taken together, this raises important questions about the existence of unviable zombie firms and their identification. Our results challenge the prevailing view that persistently weak firms are unviable, since firms can retain positive continuation value despite sustained unprofitability. Because observed outcomes are not equivalent to economic unviability, existing zombie firm measures likely

capture viable firms. More broadly, the results highlight that market fundamentals imply a 'natural' rate of zombification, and that zombie firms are not a sufficient condition for lower welfare. With standard productivity shocks playing a central role, we should expect zombie firms across countries and over the business cycle. Collectively, these results suggest that firm viability warrants greater attention in future research.

References

Acharya, V. V., Borchert, L., Jager, M., and Steffen, S. (2021a). Kicking the can down the road: Government interventions in the European banking sector. *The Review of Financial Studies*, 34(9):4090–4131.

Acharya, V. V., Crosignani, M., Eisert, T., and Eufinger, C. (2024). Zombie credit and (dis-) inflation: Evidence from Europe. *The Journal of Finance*, 79(3):1883–1929.

Acharya, V. V., Crosignani, M., Eisert, T., and Steffen, S. (2022). Zombie lending: Theoretical, international, and historical perspectives. *Annual Review of Financial Economics*, 14(1):21–38.

Acharya, V. V., Eisert, T., Eufinger, C., and Hirsch, C. (2019). Whatever it takes: The real effects of unconventional monetary policy. *The Review of Financial Studies*, 32(9):3366–3411.

Acharya, V. V., Lenzu, S., and Wang, O. (2021b). Zombie Lending and Policy Traps. *National Bureau of Economic Research*.

Adalet McGowan, M., Andrews, D., and Millot, V. (2018). The walking dead? Zombie firms and productivity performance in OECD countries. *Economic Policy*, 33(96):685–736.

Albuquerque, B. and Iyer, R. (2024). The rise of the walking dead: Zombie firms around the world. *Journal of International Economics*, 152:104019.

Albuquerque, B. and Mao, C. (2025). The zombie lending channel of monetary policy. *Management Science*.

Altman, E. I., Dai, R., and Wang, W. (2024). Global zombie companies: measurements, determinants, and outcomes. *Journal of International Business Studies*, 55(6):723–744.

Álvarez, L., García-Posada, M., and Mayordomo, S. (2023). Distressed firms, zombie firms and zombie lending: A taxonomy. *Journal of Banking & Finance*, page 106762.

Amundsen, A. (2023). Interaction effects in the adjustment cost function of firms. *Journal of Economic Dynamics and Control*, 146:104570.

Amundsen, A., Lafrance-Cooke, A., and Leung, D. (2023). Zombie firms in Canada. *Statistics Canada*.

Amundsen, A., Lafrance-Cooke, A., and Leung, D. (2025). Firm Performance, Business Supports and Zombification over the Pandemic. *International Monetary Fund*.

Amundsen, A., Leung, D., and Secrieru, O. (2026). Business dynamism and financial constraints. *Canadian Journal of Economics*, 59(1):67–105.

Andrews, D., Criscuolo, C., and Gal, P. (2016a). The global productivity slowdown, technology divergence and public policy: A firm level perspective. *Brookings Institution Hutchins Center Working Paper*, 24.

Andrews, D., Criscuolo, C., and Gal, P. N. (2016b). The best versus the rest: The global productivity slowdown, divergence across firms and the role of public policy.

Andrews, D. and Petroulakis, F. (2019). Breaking the shackles: Zombie firms, weak banks and depressed restructuring in Europe. Working Paper Series 2240, European Central Bank.

Antoshin, S., Arena, M., Gueorguiev, N., Lybek, T., Ralyea, J., and Yehoue, E. B. (2017). Credit Growth and Economic Recovery in Europe After the Global Financial Crisis. *International Monetary Fund*.

Aretz, K., Campello, M., Kankanhalli, G., and Schneider, K. (2024). Uncertainty Creates Zombie Firms: Implications for Industry Dynamics and Creative Destruction.

Autor, D., Dorn, D., Katz, L. F., Patterson, C., and Van Reenen, J. (2020). The fall of the labor share and the rise of superstar firms. *The Quarterly Journal of Economics*, 135(2):645–709.

Bajgar, M., Berlingieri, G., Calligaris, S., Criscuolo, C., and Timmis, J. (2020). Coverage and representativeness of Orbis data.

Banerjee, R. and Hofmann, B. (2018). The rise of zombie firms: Causes and consequences. *BIS Quarterly Review Spetember*.

Banerjee, R. and Hofmann, B. (2020). Corporate zombies: Anatomy and life cycle. *Economic Policy*.

Becker, B. and Ivashina, V. (2022). Weak Corporate Insolvency Rules: The Missing Driver of Zombie Lending. *AEA Papers and Proceedings*, 112:516–20.

Begenau, J. and Salomao, J. (2019). Firm financing over the business cycle. *The Review of Financial Studies*, 32(4):1235–1274.

Bittner, C., Fecht, F., and Georg, C.-P. (2021). Contagious zombies. *Available at SSRN 3829038*.

Blattner, L., Farinha, L., and Rebelo, F. (2019). When losses turn into loans: The cost of undercapitalized banks. *ECB Working paper*.

Bonfim, D., Cerqueiro, G., Degryse, H., and Ongena, S. (2022). On-site inspecting zombie lending. *Management Science*.

Bordo, M. D., Redish, A., and Rockoff, H. (2015). Why didn't Canada have a banking crisis in 2008 (or in 1930, or 1907, or...)? *The Economic History Review*, 68(1):218–243.

Bruche, M. and Llobet, G. (2014). Preventing zombie lending. *The Review of Financial Studies*, 27(3):923–956.

Brunner, M., De Martiis, A., and Valta, P. (2024). How does competition affect zombie firms? *Available at SSRN 4310399*.

Caballero, R. J., Hoshi, T., and Kashyap, A. K. (2008). Zombie lending and depressed restructuring in Japan. *American Economic Review*, 98(5):1943–77.

Carreira, C., Teixeira, P., and Nieto-Carrillo, E. (2022). Recovery and exit of zombie firms in Portugal. *Small Business Economics*, 59(2):491–519.

Cella, C. (2020). Zombie firms in Sweden: Implications for the real economy and financial stability. *Staff Memo, Sveriges Riksbank*.

Chari, A., Jain, L., and Kulkarni, N. (2021). The unholy trinity: Regulatory forbearance, stressed banks and zombie firms. Technical report, National Bureau of Economic Research.

Chen, Y. and Tombe, T. (2026). Aggregate productivity effect of labour and capital market distortions in Canada. *Canadian Journal of Economics*, 59(1):42–66.

Choi, J., Levchenko, A. A., Ruzic, D., and Shim, Y. (2024). Superstars or supervillains? Large firms in the South Korean growth miracle. Technical report, National Bureau of Economic Research.

Chopra, Y., Subramanian, K., and Tantri, P. L. (2021). Bank cleanups, capitalization, and lending: Evidence from India. *The Review of Financial Studies*, 34(9):4132–4176.

Clementi, G. L. and Palazzo, B. (2016). Entry, exit, firm dynamics, and aggregate fluctuations. *American Economic Journal: Macroeconomics*, 8(3):1–41.

Collard-Wexler, A. and De Loecker, J. (2021). Productivity and Capital Measurement Error. *NBER Working Paper*.

De Jonghe, O., Mulier, K., and Samarin, I. (2021). Bank specialization and zombie lending. Technical report, NBB Working Paper.

De Martiis, A. and Peter, F. J. (2021). When companies don't die: Analyzing zombie and distressed firms in a low interest rate environment. *Available at SSRN 3890788*.

Decker, R., Haltiwanger, J., Jarmin, R., and Miranda, J. (2014). The role of entrepreneurship in U.S. job creation and economic dynamism. *Journal of Economic Perspectives*, 28(3):3–24.

Díez, F. J., Fan, J., and Villegas-Sánchez, C. (2021). Global declining competition? *Journal of International Economics*, 132:103492.

Dunne, T., Roberts, M. J., and Samuelson, L. (1989). The Growth and Failure of U.S. Manufacturing Plants. *The Quarterly Journal of Economics*, 104(4):671–698.

El Ghoul, S., Fu, Z., and Guedhami, O. (2021). Zombie firms: Prevalence, determinants, and corporate policies. *Finance Research Letters*, 41:101876.

Faria-e-Castro, M., Paul, P., and Sánchez, J. M. (2024). Evergreening. *Journal of Financial Economics*, 153:103778.

Favara, G., Minoiu, C., and Perez-Orive, A. (2024). Zombie Lending to U.S. Firms. FRB Atlanta Working Paper 2024-7, Federal Reserve Bank of Atlanta.

Foster, L., Haltiwanger, J., and Syverson, C. (2008). Reallocation, Firm Turnover, and Efficiency: Selection on Productivity or Profitability? *American Economic Review*, 98(1):394–425.

Giannetti, M. and Simonov, A. (2013). On the real effects of bank bailouts: Micro evidence from Japan. *American Economic Journal: Macroeconomics*, 5(1):135–67.

Gopinath, G., Kalemli-Özcan, Ş., Karabarbounis, L., and Villegas-Sanchez, C. (2017). Capital allocation and productivity in South Europe. *The Quarterly Journal of Economics*, 132(4):1915–1967.

Gouveia, A. F. and Osterhold, C. (2018). Fear the walking dead: Zombie firms, spillovers and exit barriers.

Grieder, T., Khan, M., Ortega, J., and Symmers, C. (2021). COVID-19's impact on the financial health of Canadian businesses: An initial assessment. Technical report, Bank of Canada.

Grieder, T. and Ortega, J. (2020). A financial stability analysis of zombie firms in Canada. Technical report, Bank of Canada.

Griliches, Z. and Regev, H. (1995). Firm productivity in Israeli industry 1979–1988. *Journal of Econometrics*, 65(1):175–203.

Gu, W. (2019). Frontier firms, productivity dispersion and aggregate productivity growth in Canada. *International Productivity Monitor*, 37:96–119.

Hamano, M., Schnattinger, P., Shintani, M., Uesugi, I., and Zanetti, F. (2025). Credit market tightness and zombie firms: theory and evidence. Technical report, CESifo.

Havemeister, L. K. and Horn, K. (2023). Fear the walking dead? Zombie firms in the euro area and their effect on healthy firms' credit conditions. *ESRB Working Paper Series*, (143).

Hennessy, C. A. and Whited, T. M. (2007). How costly is external financing? Evidence from a structural estimation. *The Journal of Finance*, 62(4):1705–1745.

Herweg, F. and Kähny, M. (2022). Do zombies rise when interest rates fall? A relationship banking model.

Hong, M. G. H., Igan, M. D. O., and Lee, D. (2021). Zombies on the brink: Evidence from Japan on the reversal of monetary policy effectiveness.

Hopenhayn, H. and Rogerson, R. (1993). Job turnover and policy evaluation: A general equilibrium analysis. *Journal of Political Economy*, 101(5):915–938.

Hopenhayn, H. A. (1992). Entry, exit, and firm dynamics in long run equilibrium. *Econometrica: Journal of the Econometric Society*, pages 1127–1150.

Hoshi, T. (2006). Economics of the living dead. *The Japanese Economic Review*, 57(1):30–49.

Hoshi, T., Kawaguchi, D., and Ueda, K. (2023). Zombies, again? The COVID-19 business support programs in Japan. *Journal of Banking & Finance*, 147:106421.

Hu, Y. and Varas, F. (2021). A theory of zombie lending. *The Journal of Finance*, 76(4):1813–1867.

Jordà, Ò., Kornejew, M., Schularick, M., and Taylor, A. M. (2022). Zombies at large? Corporate debt overhang and the macroeconomy. *The Review of Financial Studies*, 35(10):4561–4586.

Kalemli-Özcan, Ş., Sørensen, B. E., Villegas-Sanchez, C., Volosovych, V., and Yeşiltaş, S. (2024). How to construct nationally representative firm-level data from the Orbis global database: New facts on SMEs and aggregate implications for industry concentration. *American Economic Journal: Macroeconomics*, 16(2):353–374.

Kirkby, R. (2017). A toolkit for value function iteration. *Computational Economics*, 49(1):1–15.

Kopecky, K. A. and Suen, R. M. (2010). Finite state Markov-chain approximations to highly persistent processes. *Review of Economic Dynamics*, 13(3):701–714.

Kulkarni, N., Ritadhi, S., Vij, S., and Waldock, K. (2025). Unearthing zombies. *Management Science*.

Kwon, H. U., Narita, F., and Narita, M. (2015). Resource reallocation and zombie lending in Japan in the 1990s. *Review of Economic Dynamics*, 18(4):709–732.

Lam, W. R., Schipke, M. A., Tan, Y., and Tan, Z. (2017). Resolving China's zombies: Tackling debt and raising productivity.

Li, B. and Ponticelli, J. (2022). Going bankrupt in China. *Review of Finance*, 26(3):449–486.

McGowan, M. A., Andrews, D., and Millot, V. (2017). Insolvency regimes, zombie firms and capital reallocation.

Midrigan, V. and Xu, D. Y. (2014). Finance and misallocation: Evidence from plant-level data. *American Economic Review*, 104(2):422–458.

Mingarelli, L., Ravanetti, B., Shakir, T., and Wendelborn, J. (2022). Dawn of the (half) dead: The twisted world of zombie identification.

Nurmi, S., Vanhala, J., and Virén, M. (2020). The life and death of zombies - Evidence from government subsidies to firms. *Bank of Finland Research Discussion Paper*, (8).

Ollivaud, P., Guillemette, Y., and Turner, D. (2016). Links between weak investment and the slowdown in productivity and potential output growth across the OECD.

Osotimehin, S. and Pappadà, F. (2017). Credit frictions and the cleansing effect of recessions. *The Economic Journal*, 127(602):1153–1187.

Ozdagli, A. K. and Wang, J. (2022). Uncertainty, stock prices and debt structure: Evidence from the U.S.-China trade war.

Passalacqua, A., Angelini, P., Lotti, F., and Soggia, G. (2021). The real effects of bank supervision: Evidence from on-site bank inspections. *Bank of Italy Temi di Discussione (Working Paper) No*, 1349.

Peek, J. and Rosengren, E. S. (2005). Unnatural selection: Perverse incentives and the misallocation of credit in Japan. *American Economic Review*, 95(4):1144–1166.

Qiao, L. and Fei, J. (2022). Government subsidies, enterprise operating efficiency, and 'stiff but deathless' zombie firms. *Economic Modelling*, 107:105728.

Ren, M., Zhao, J., and Zhao, J. (2023). The crowding-out effect of zombie companies on fixed asset investment: Evidence from China. *Research in International Business and Finance*, 65:101979.

Schivardi, F., Sette, E., and Tabellini, G. (2022). Credit misallocation during the European financial crisis. *The Economic Journal*, 132(641):391–423.

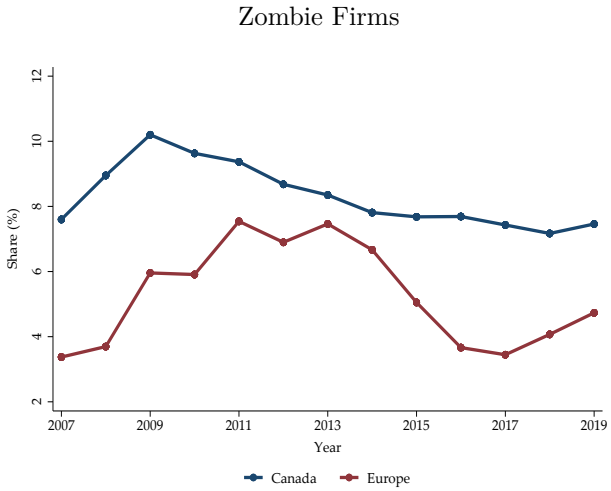
Schmidt, C., Schneider, Y., Steffen, S., and Streitz, D. (2020). Capital misallocation and innovation. *Available at SSRN 3489801*.

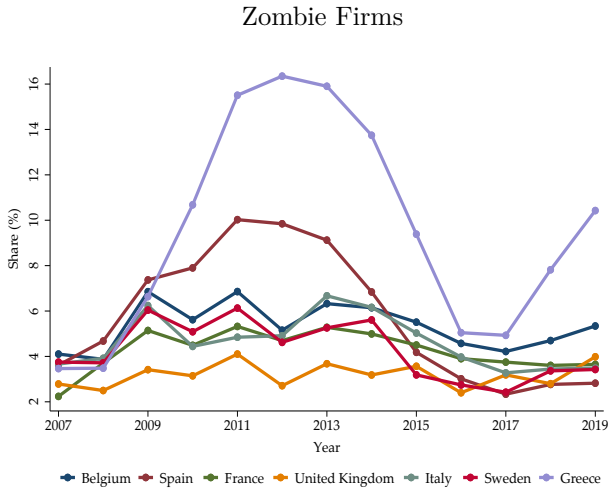
Sharpe, A. and Tsang, J. (2018). The stylized facts about slower productivity growth in Canada. *International Productivity Monitor*, (35):52–72.

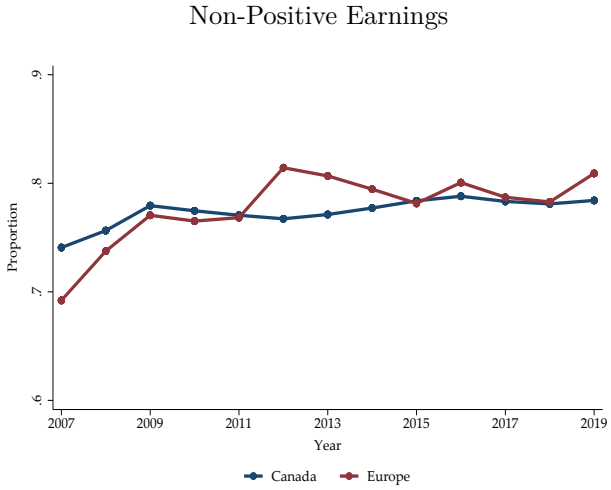
Storz, M., Koetter, M., Setzer, R., and Westphal, A. (2017). Do we want these two to tango? On zombie firms and stressed banks in Europe.

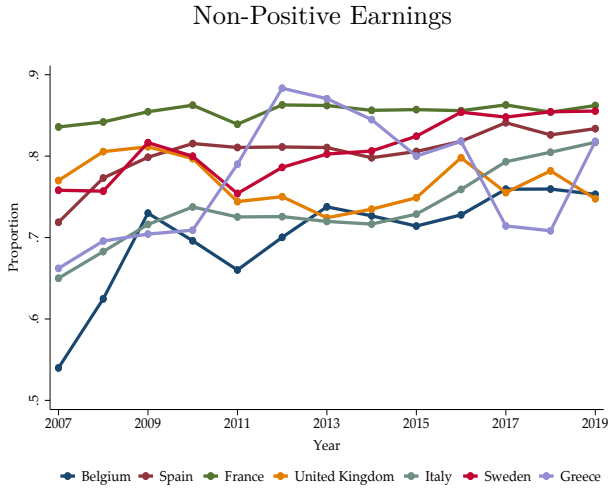
Tracey, B. (2025). The real effects of zombie lending in Europe. *Oxford Bulletin of Economics and Statistics*, 87(1):122–154.

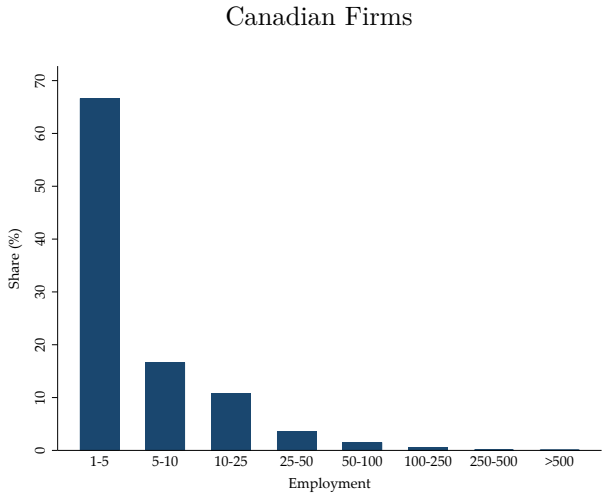
A Favara et al. (2024) Zombie Firm Definition

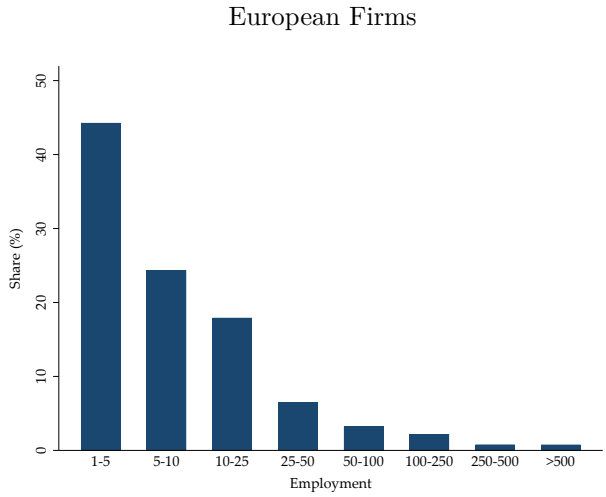












Canadian Firms

	Healthy (%)	Zombie (%)
Healthy (%)	94.54	5.46
Zombie (%)	45.45	54.55

European Firms

	Healthy (%)	Zombie (%)
Healthy (%)	95.88	4.12
Zombie (%)	51.99	48.01

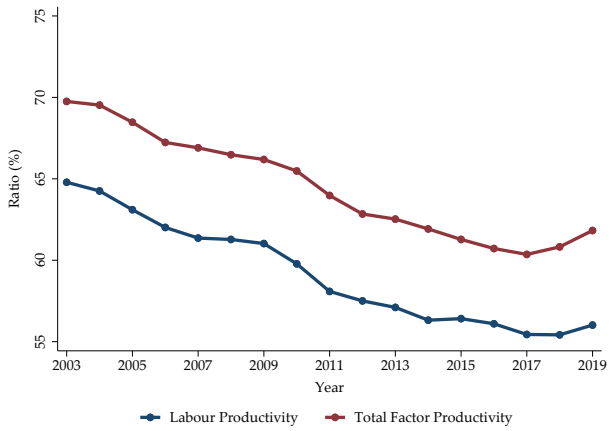
Canadian Firms

	Number of Zombie Years (%)	Duration of Zombie Spells (%)	Number of Zombie Spells (%)
1	41.81	58.59	70.89
2	23.38	23.36	21.22
3	13.81	10.04	7.89
4	8.51	4.31	.
5	5.18	1.89	.
6	3.09	0.90	.
7	1.87	0.45	.
8	1.09	0.22	.
9	0.59	0.12	.
10	0.32	0.06	.
>10	0.35	0.07	.

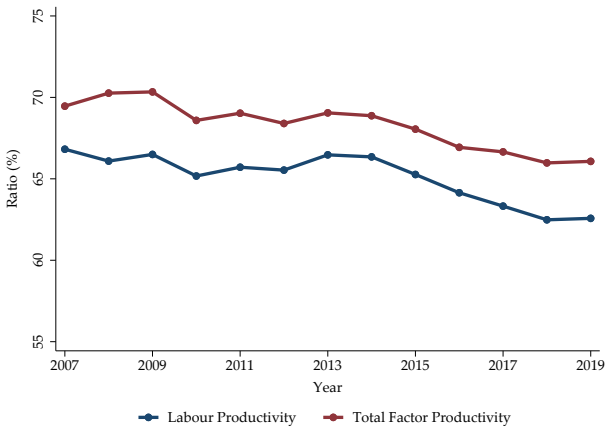
European Firms

	Number of Zombie Years (%)	Duration of Zombie Spells (%)	Number of Zombie Spells (%)
1	54.84	68.09	81.45
2	23.64	19.65	15.73
3	11.36	7.50	2.82
4	5.38	2.75	.
5	2.59	1.14	.
6	1.23	0.49	.
7	0.55	0.22	.
8	0.25	0.09	.
9	0.10	0.04	.
10	0.05	0.02	.
>10	0.03	0.01	.

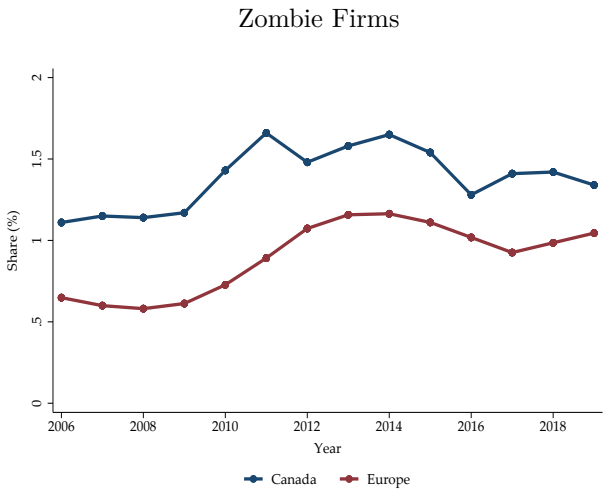
Canadian Firms

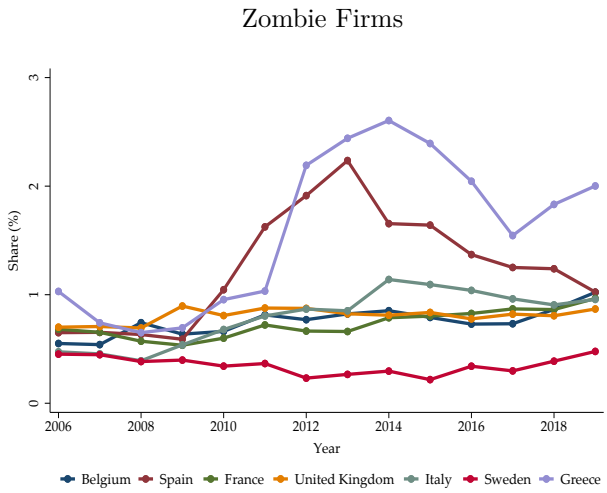


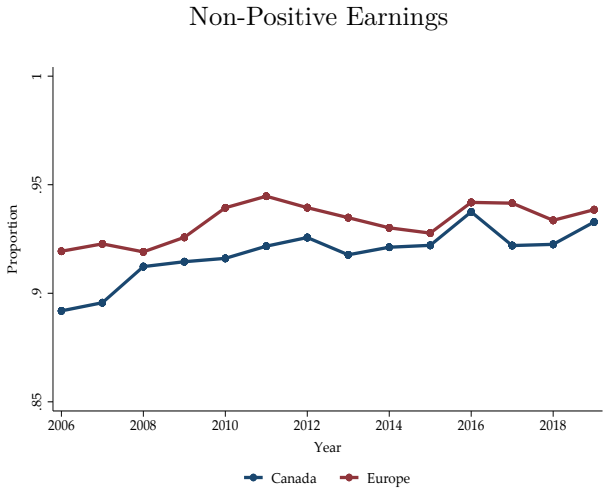
European Firms

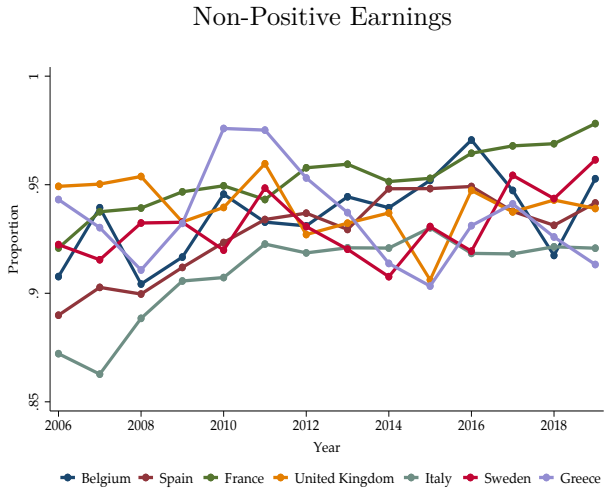


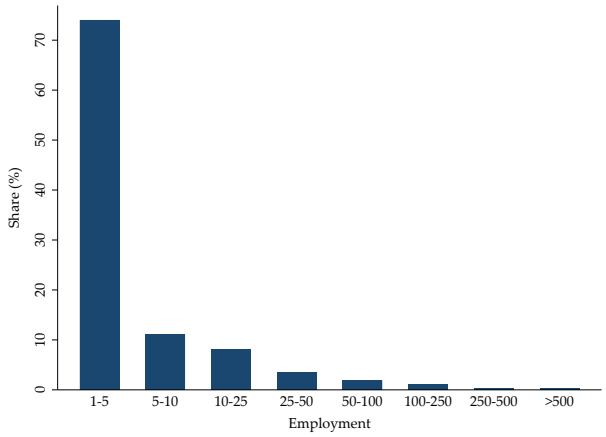
B Subsidized-Credit Zombie Firm Definition

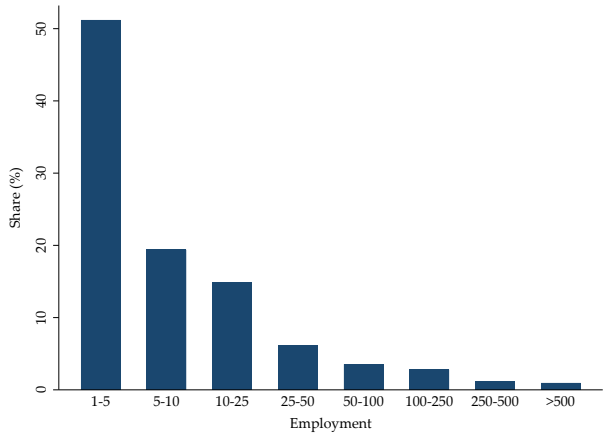












Canadian Firms

	Healthy (%)	Zombie (%)
Healthy (%)	99.27	0.73
Zombie (%)	36.11	63.89

European Firms

	Healthy (%)	Zombie (%)
Healthy (%)	99.35	0.65
Zombie (%)	42.11	57.89

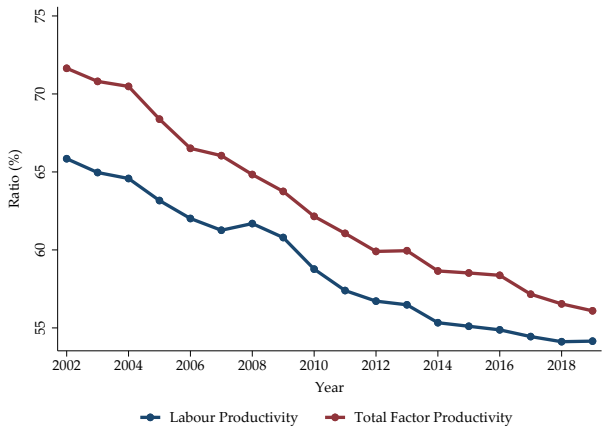
Canadian Firms

	Number of Zombie Years (%)	Duration of Zombie Spells (%)	Number of Zombie Spells (%)
1	48.31	54.27	88.45
2	22.43	22.23	10.43
3	11.66	10.34	1.12
4	6.58	5.25	.
5	3.94	3.00	.
6	2.42	1.74	.
7	1.47	1.09	.
8	1.02	0.66	.
9	0.71	0.49	.
10	0.47	0.29	.
>10	0.98	0.65	.

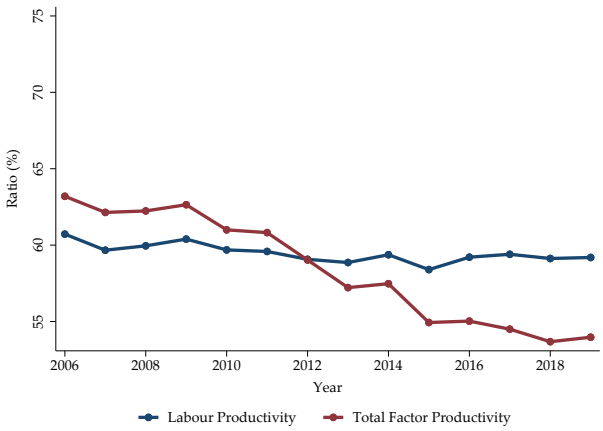
European Firms

	Number of Zombie Years (%)	Duration of Zombie Spells (%)	Number of Zombie Spells (%)
1	59.83	64.83	91.63
2	21.44	20.35	7.70
3	9.28	7.89	0.67
4	4.55	3.50	.
5	2.20	1.59	.
6	1.19	0.86	.
7	0.68	0.43	.
8	0.38	0.24	.
9	0.21	0.13	.
10	0.10	0.07	.
>10	0.14	0.09	.

Canadian Firms



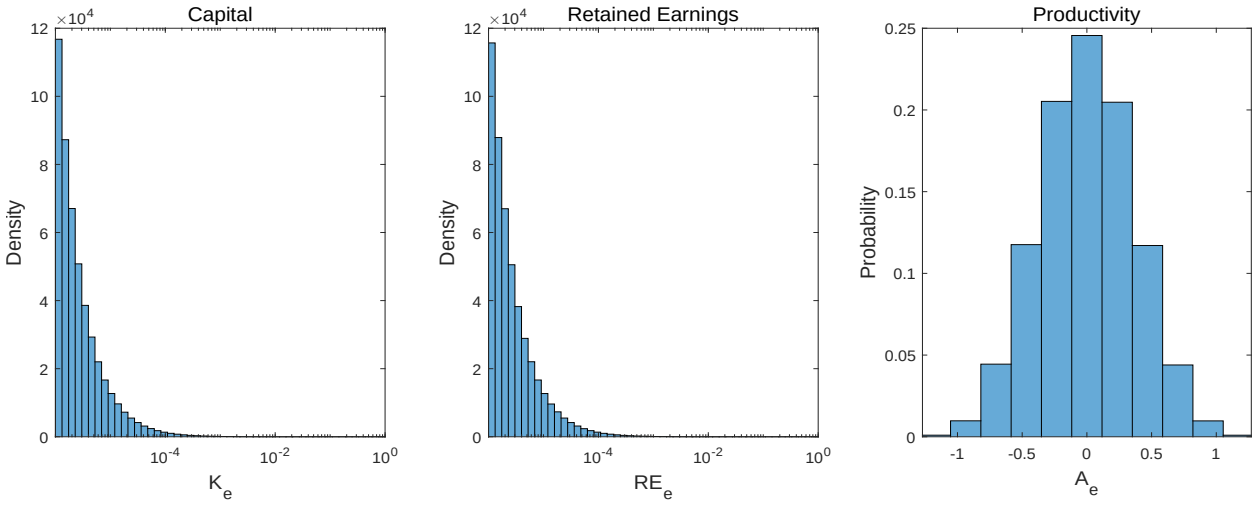
European Firms



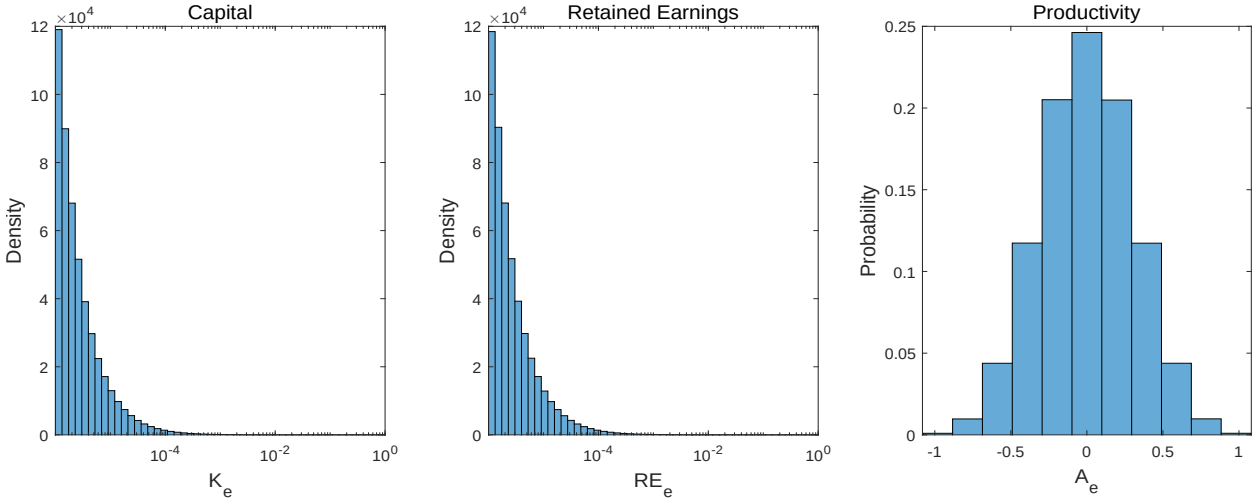
C Model Dynamics

C.1 Ex-Ante Entrant Distributions

Canadian Firms

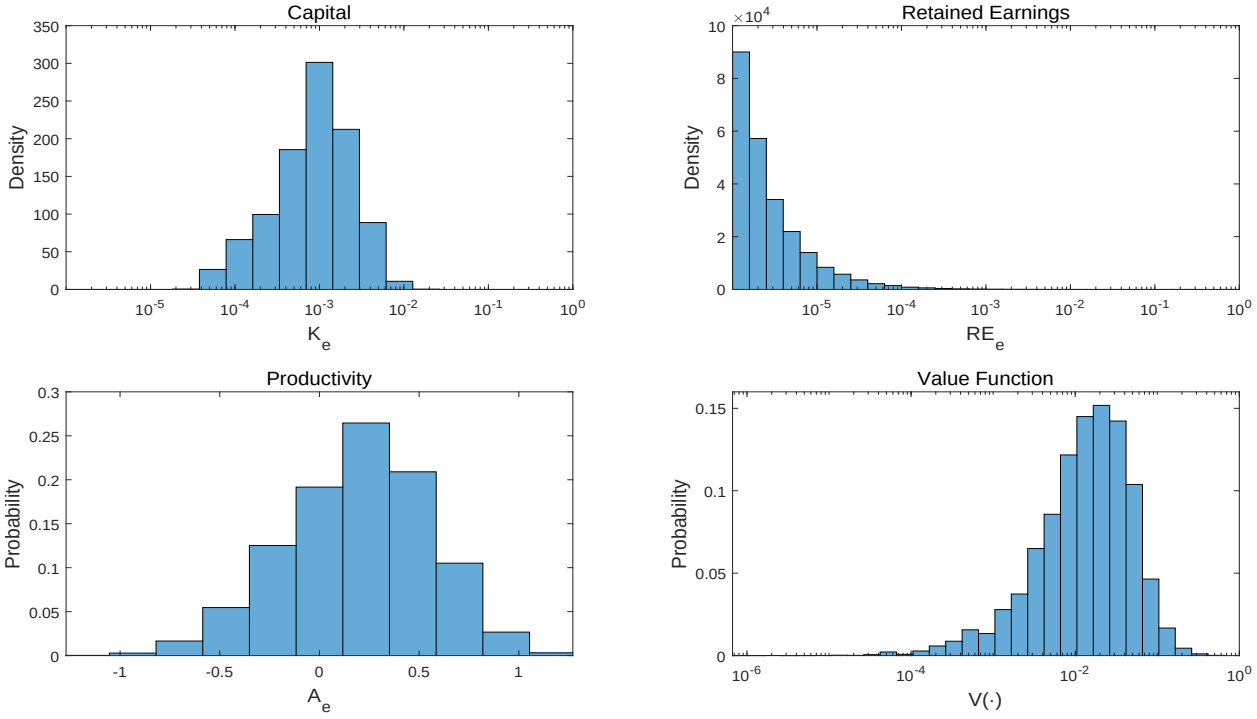


European Firms

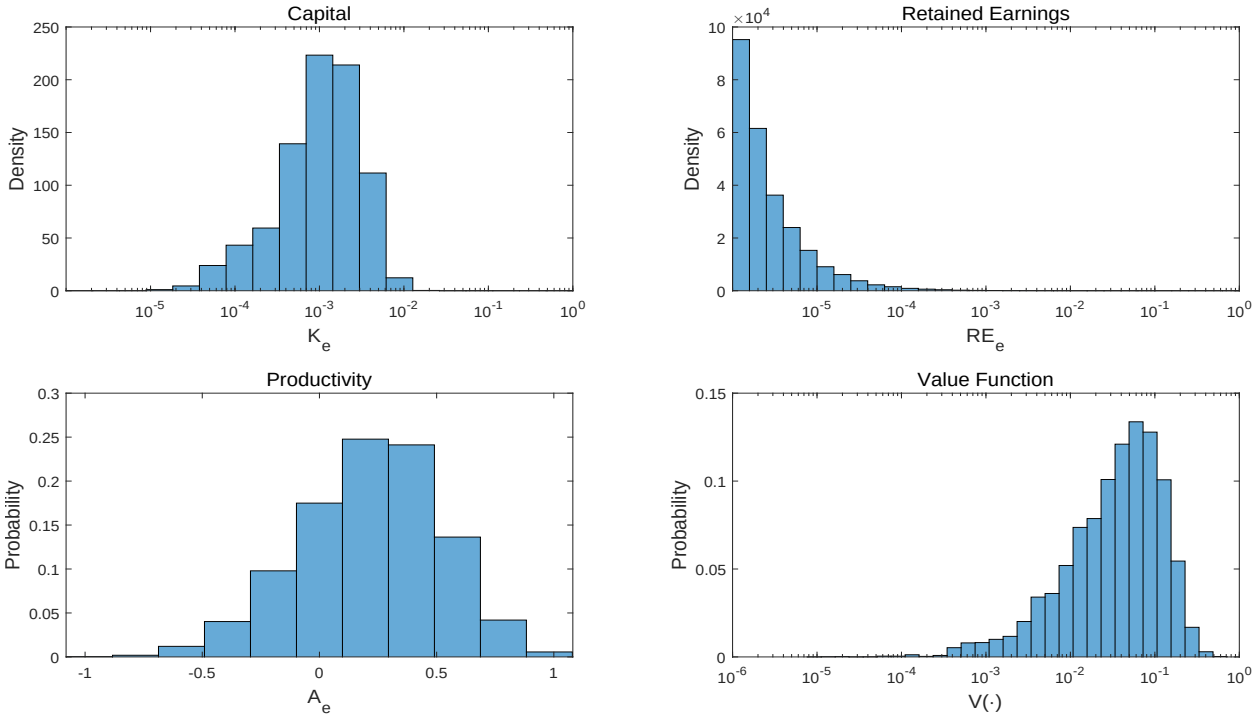


C.2 Ex-Post Entrant Distributions

Canadian Firms

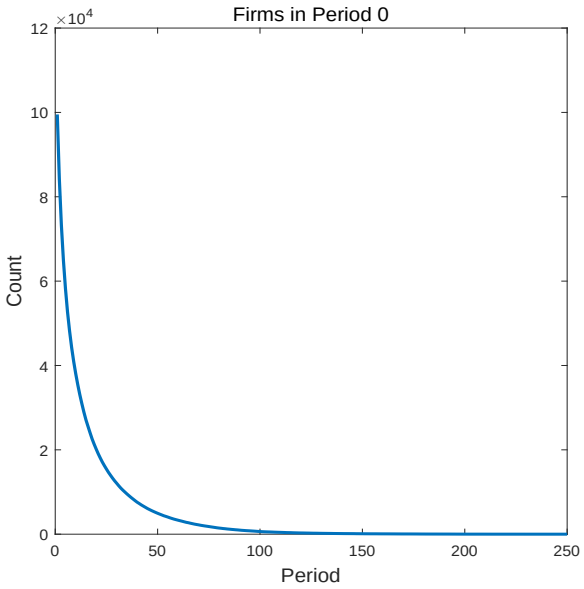


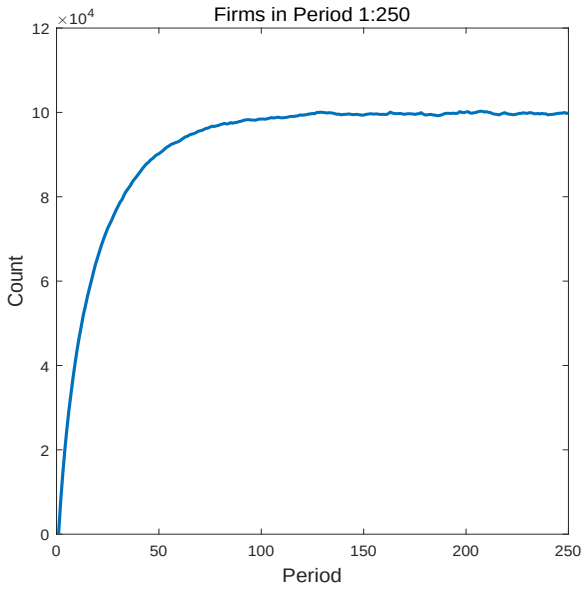
European Firms



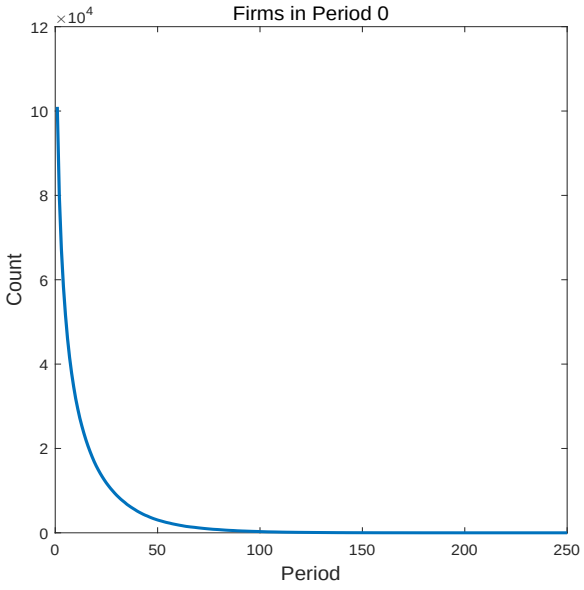
C.3 Count of Firms

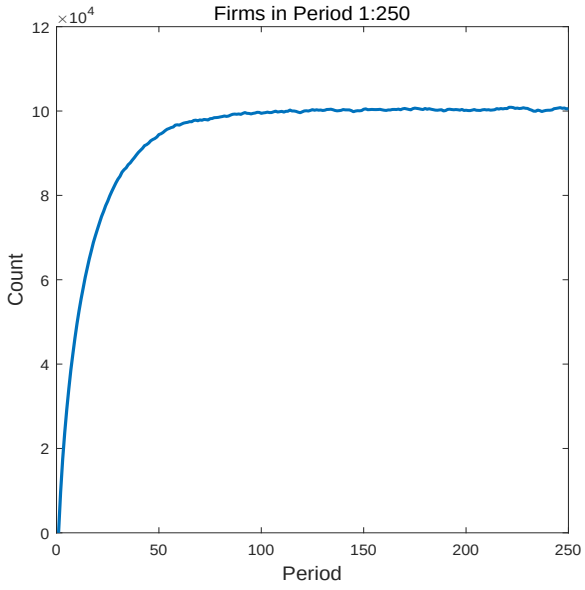
Canadian Firms





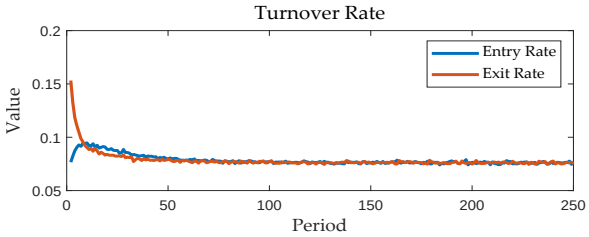
European Firms

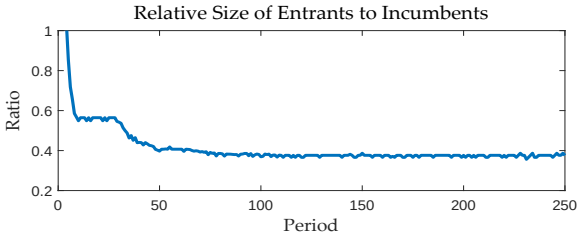


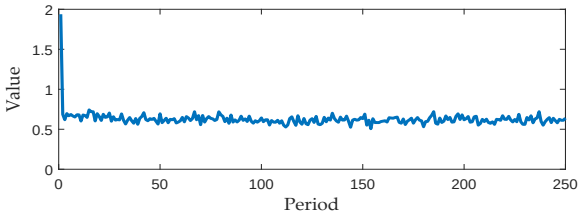


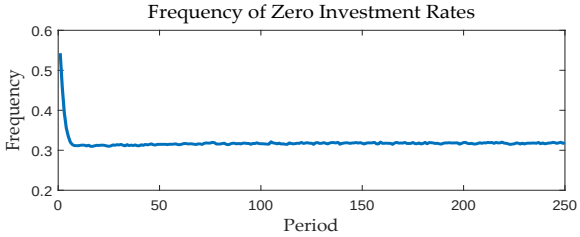
C.4 Moments

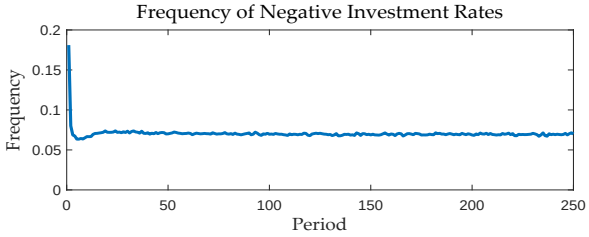
Canadian Firms

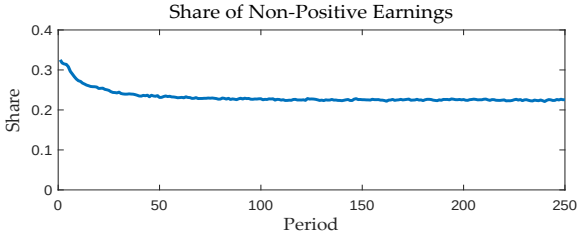




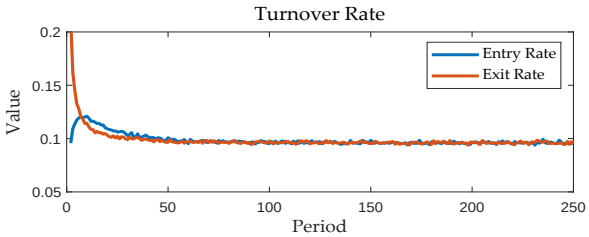


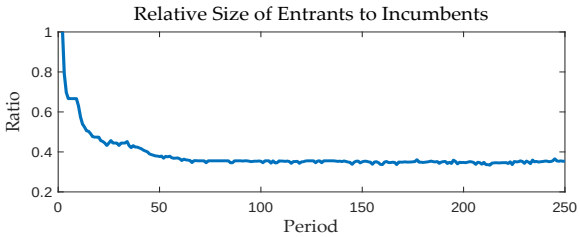


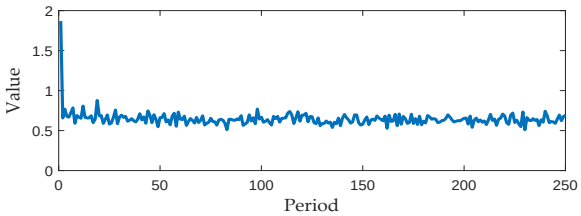


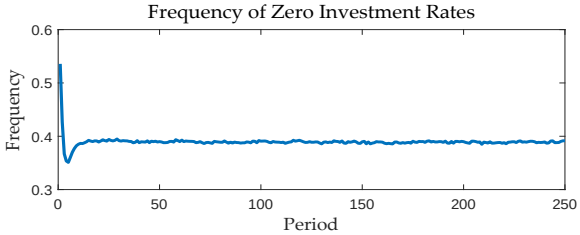


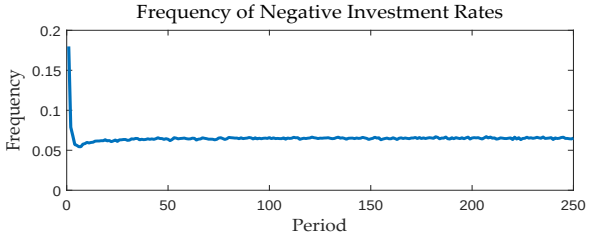
European Firms

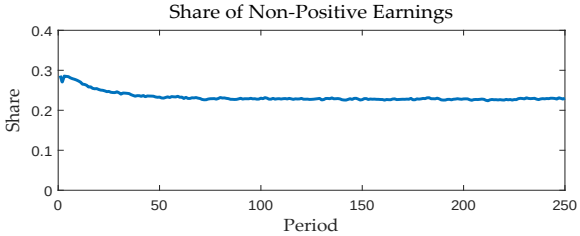






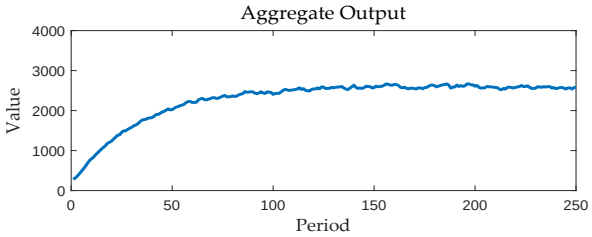


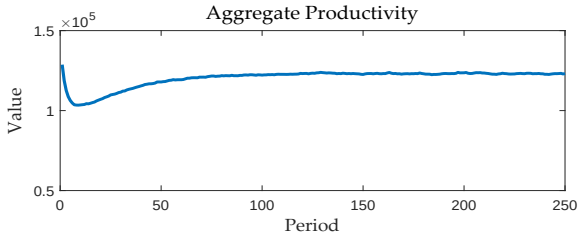


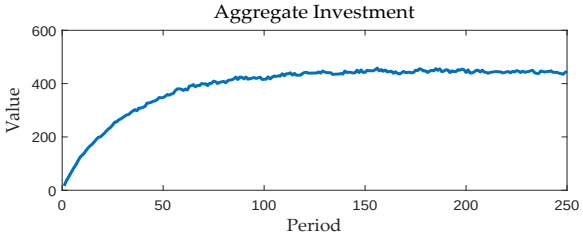


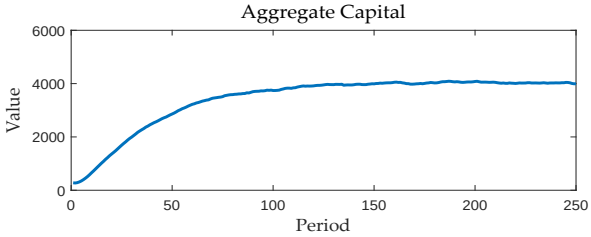
C.5 Variables

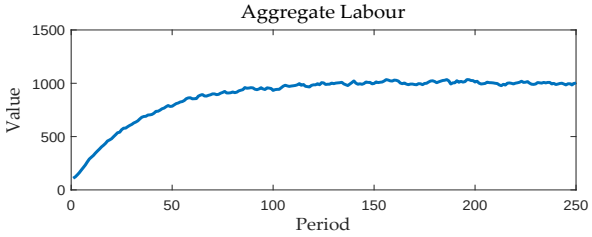
Canadian Firms

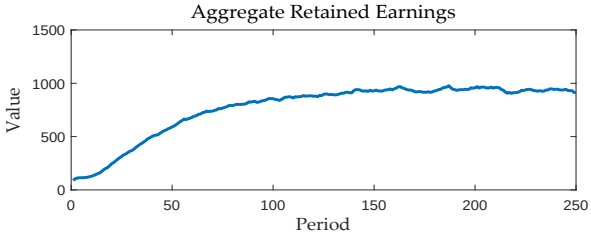




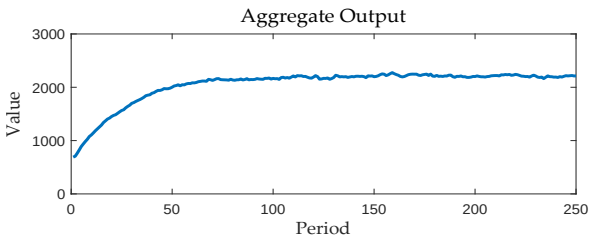


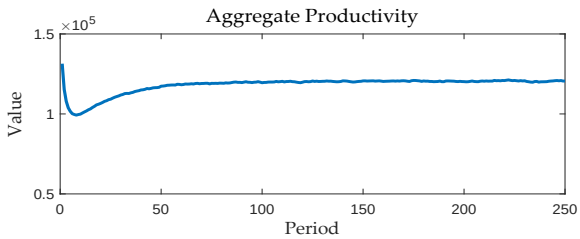


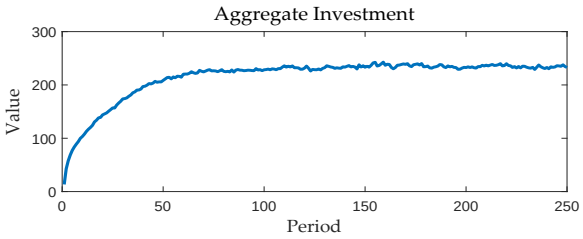


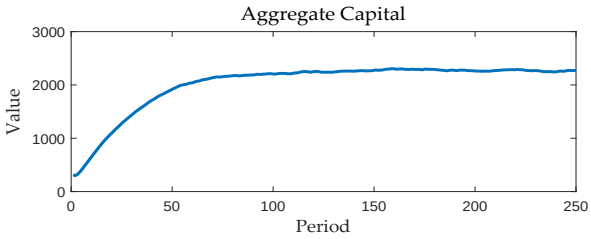


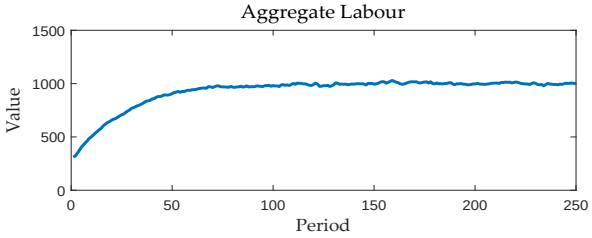
European Firms

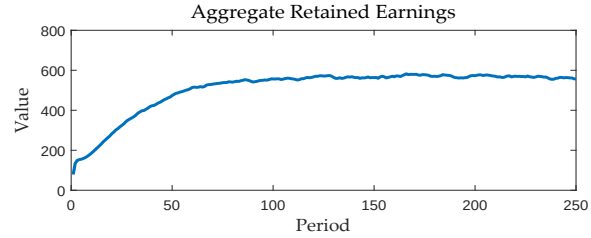






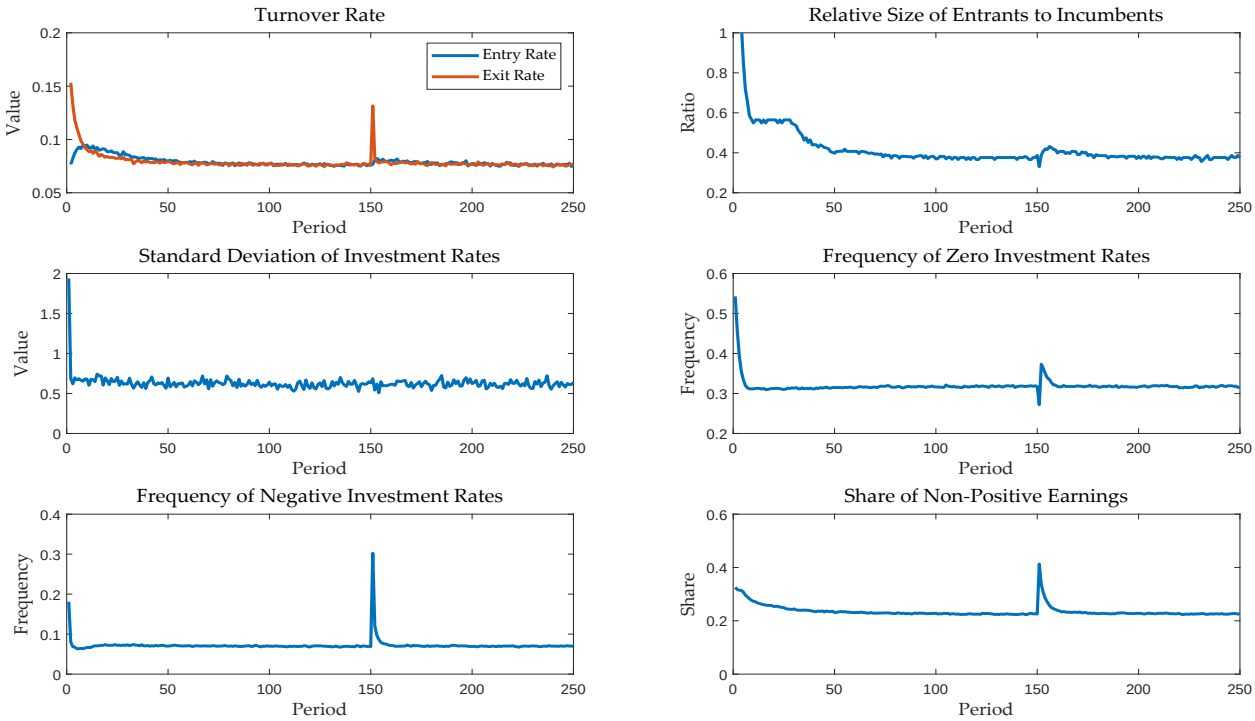




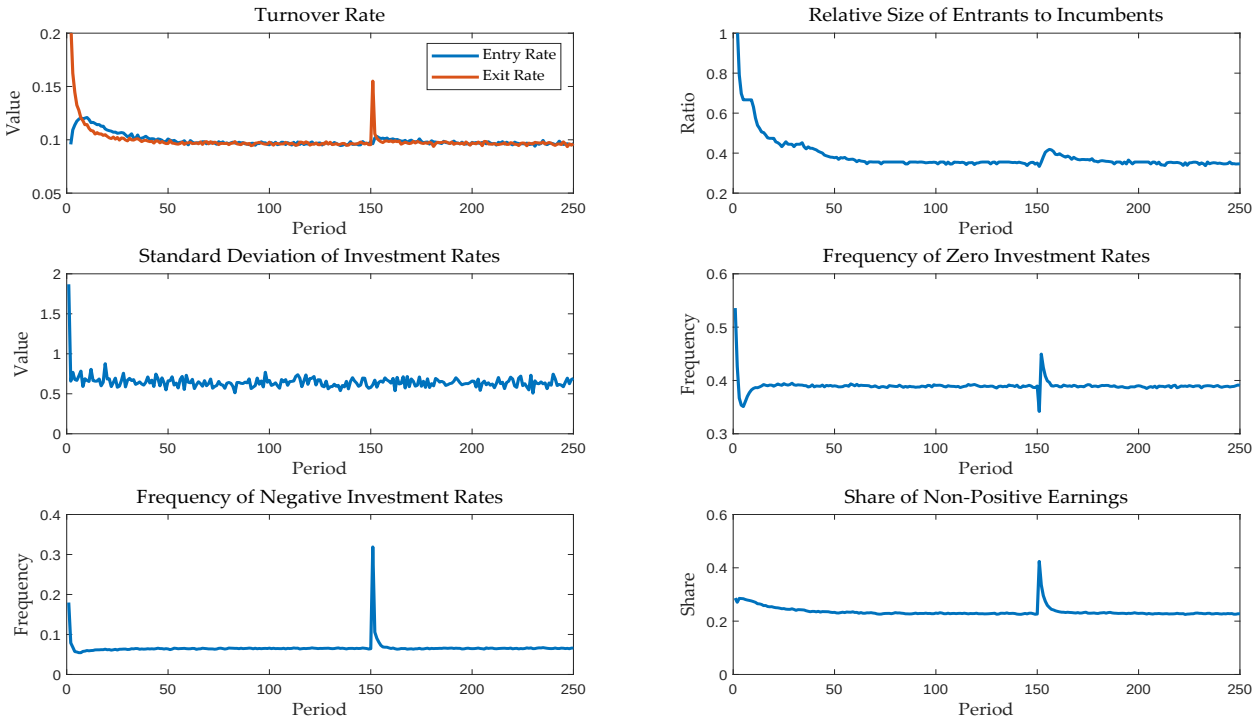


C.6 Counterfactual: Aggregate Shock – Moments

Canadian Firms

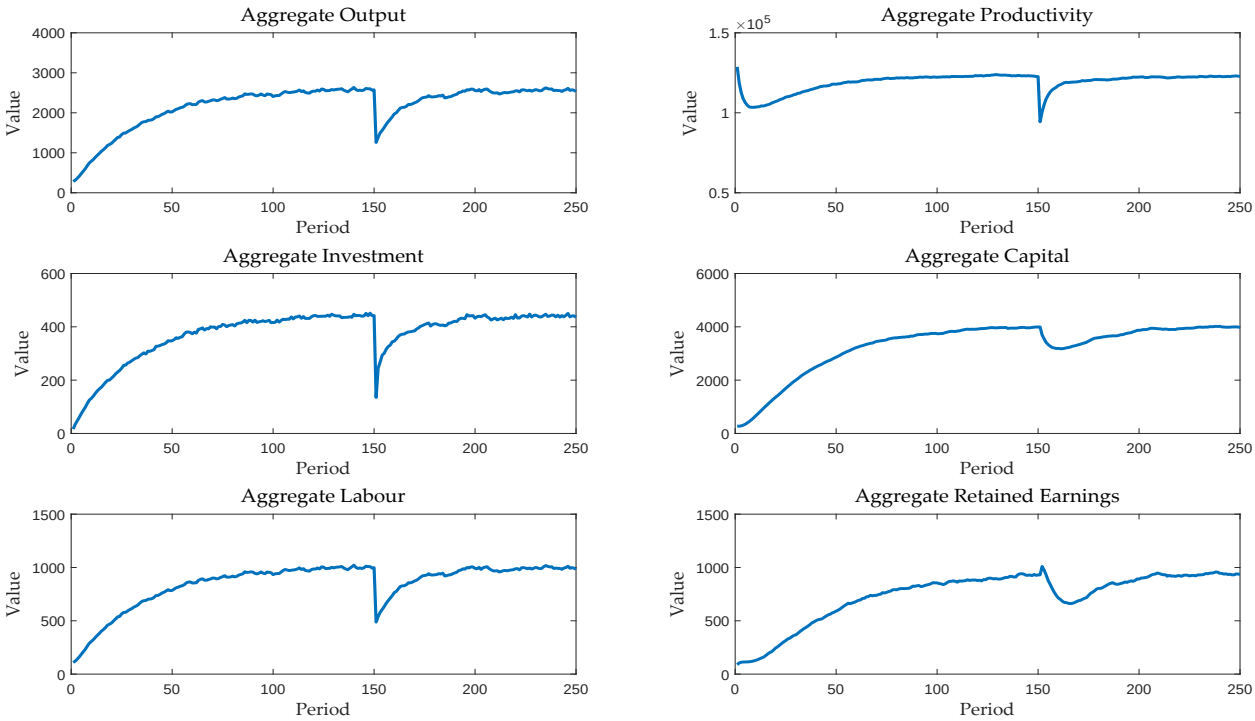


European Firms

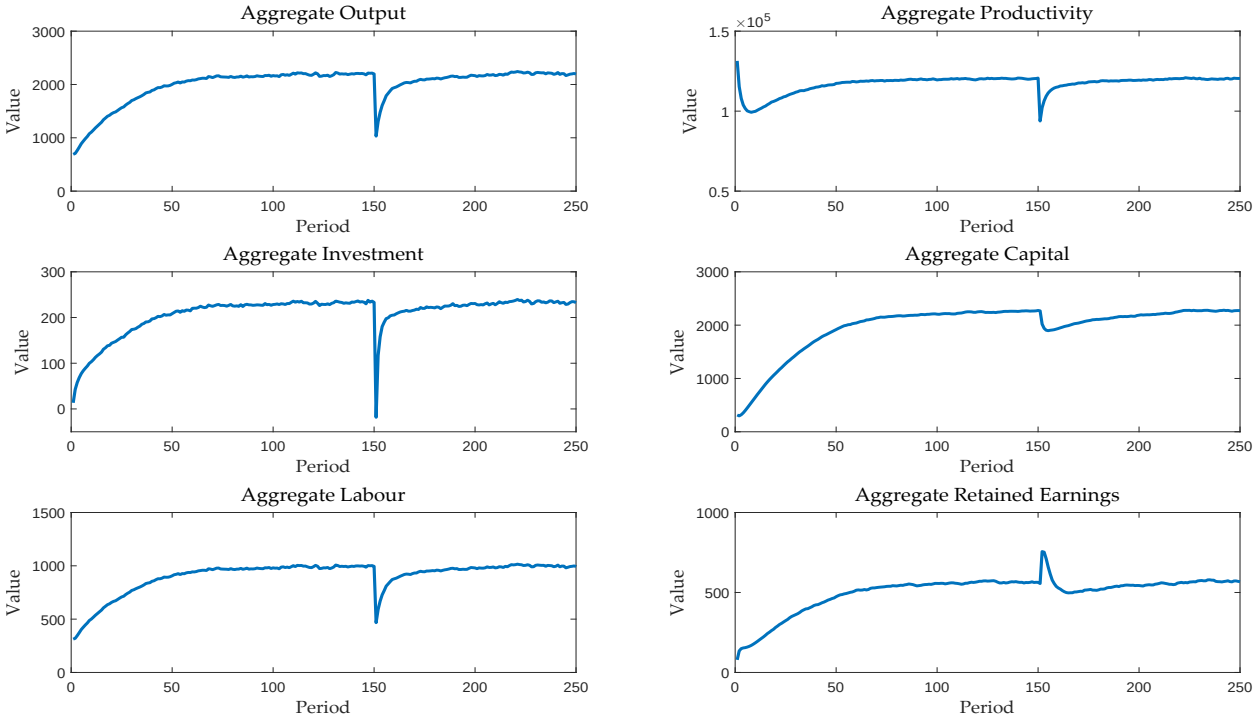


C.7 Counterfactual: Aggregate Shock – Variables

Canadian Firms

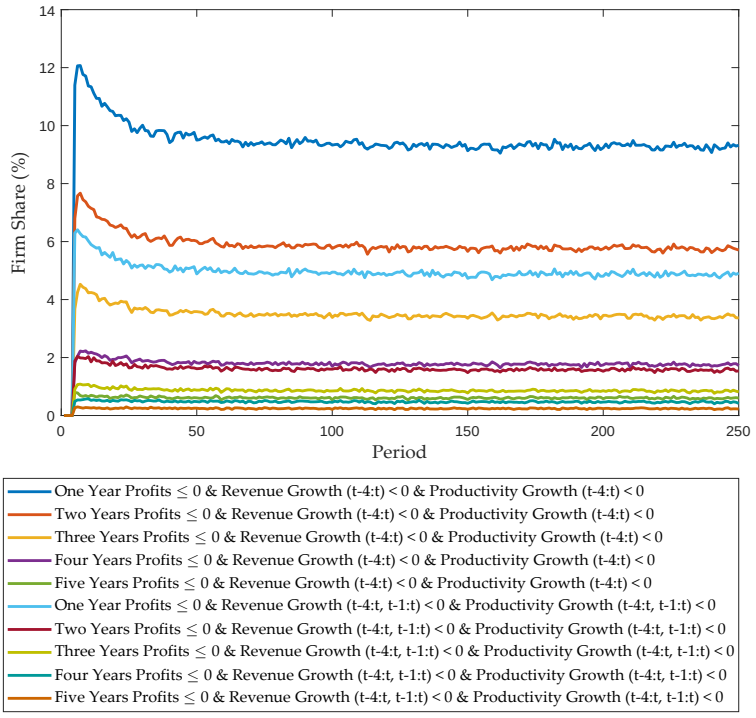


European Firms

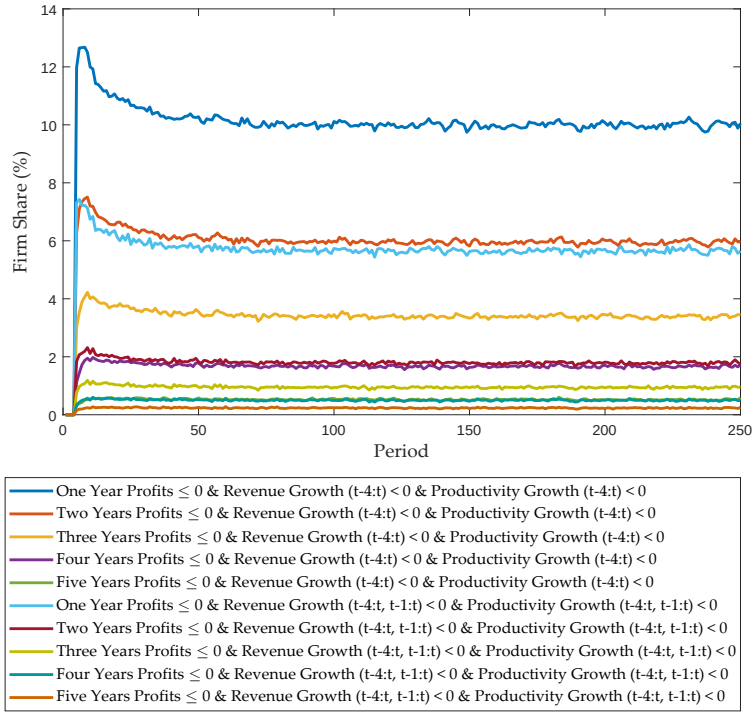


C.8 Unviable Zombie Firm Definition: Per Consecutive Year

Canadian Firms

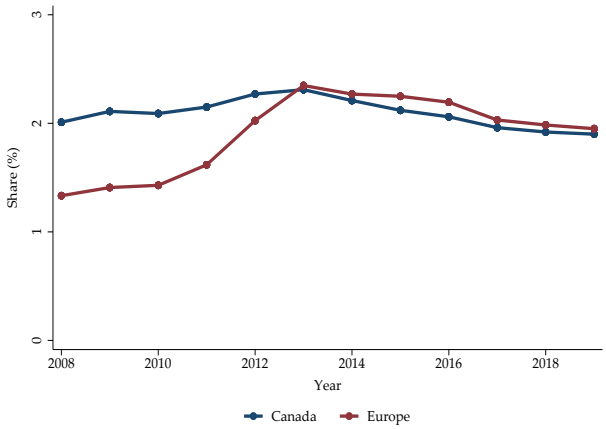


European Firms

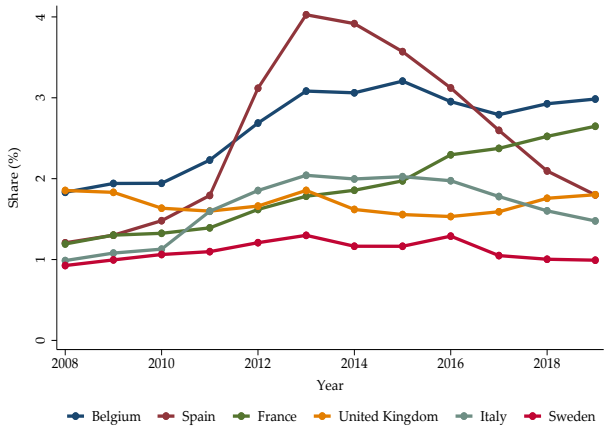


C.9 Unviable Zombie Firm Definition: Components

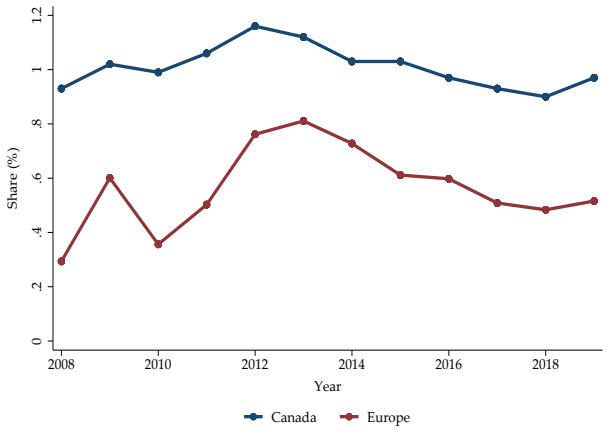
Earnings Criteria



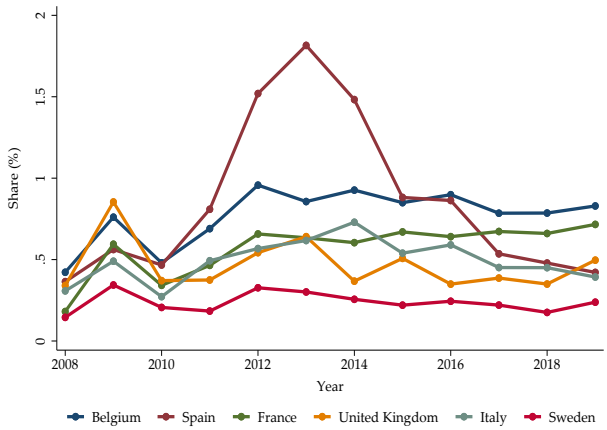
Earnings Criteria



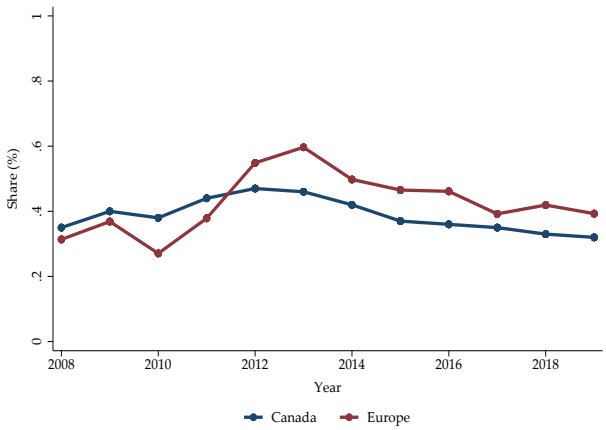
Earnings and Revenue Criteria



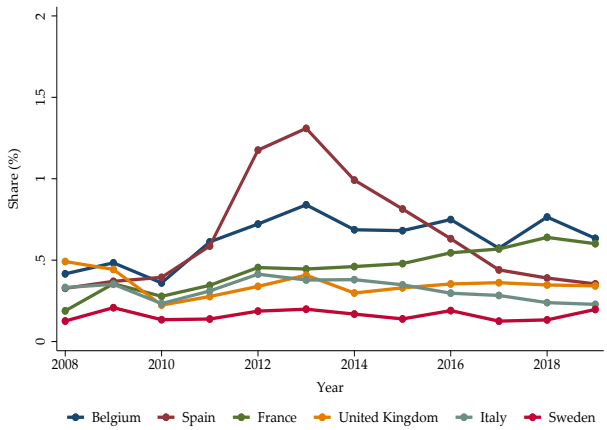
Earnings and Revenue Criteria



Earnings and Productivity Criteria



Earnings and Productivity Criteria





PUBLICATIONS

Rethinking Zombie Firms: From Volatility to Viability

Working Paper No. WP/2026/227